Support Vector Machines (examples)

Machine Learning Techniques

Karthik Thiagarajan

Outline

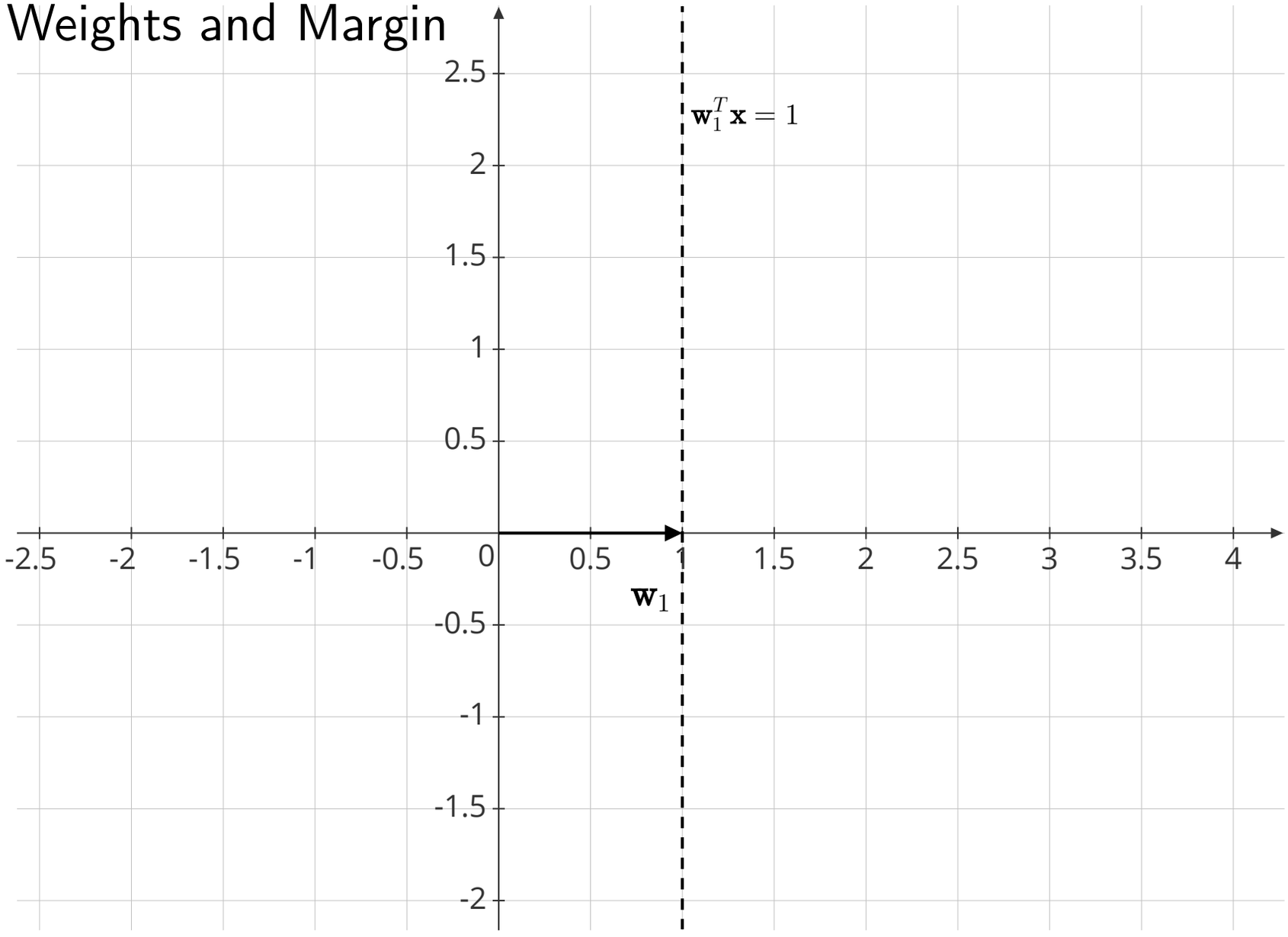
Primal Picture

- Weights and Margin
- Feature space
- Parameter space
- Constraints
- Objective
- Interaction between objective and constraints
- Active and inactive constraints

Dual Picture

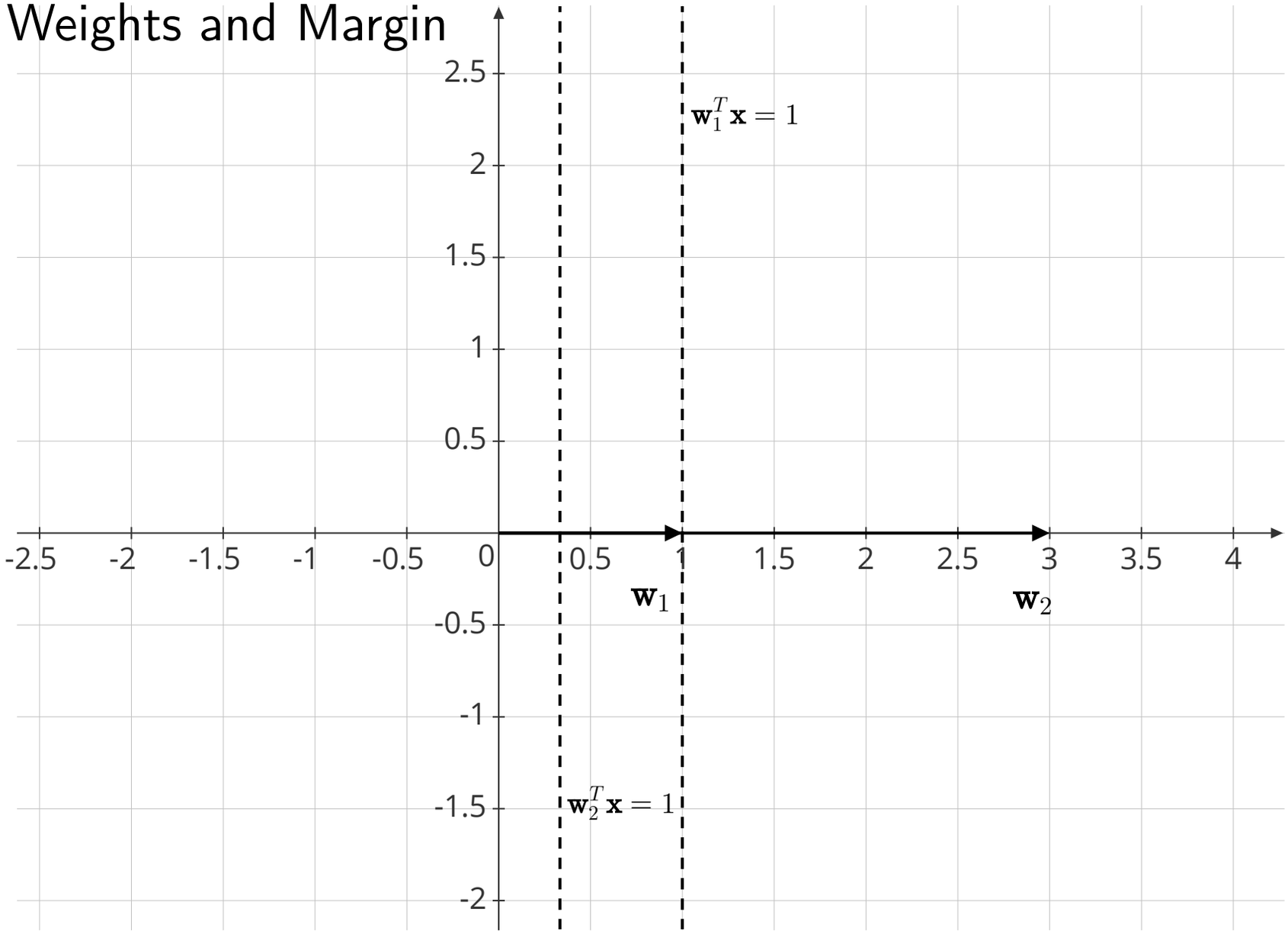
- Lagrange multipliers
- Weight vector

Weights and Margin



$$\mathbf{w}_1 = \begin{bmatrix} 1 & 0 \end{bmatrix}^T$$

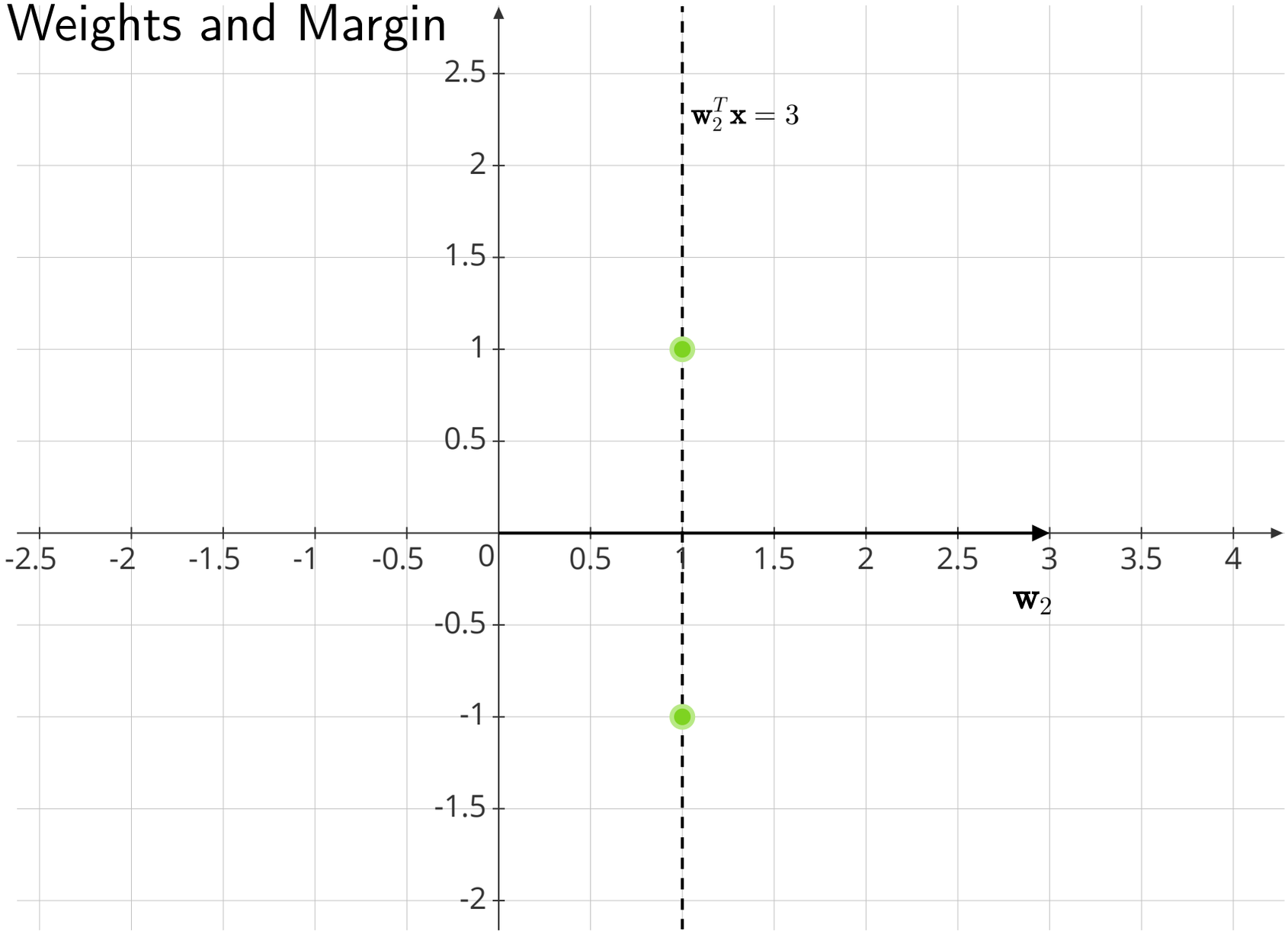
Weights and Margin



$$\mathbf{w}_1 = \begin{bmatrix} 1 & 0 \end{bmatrix}^T$$

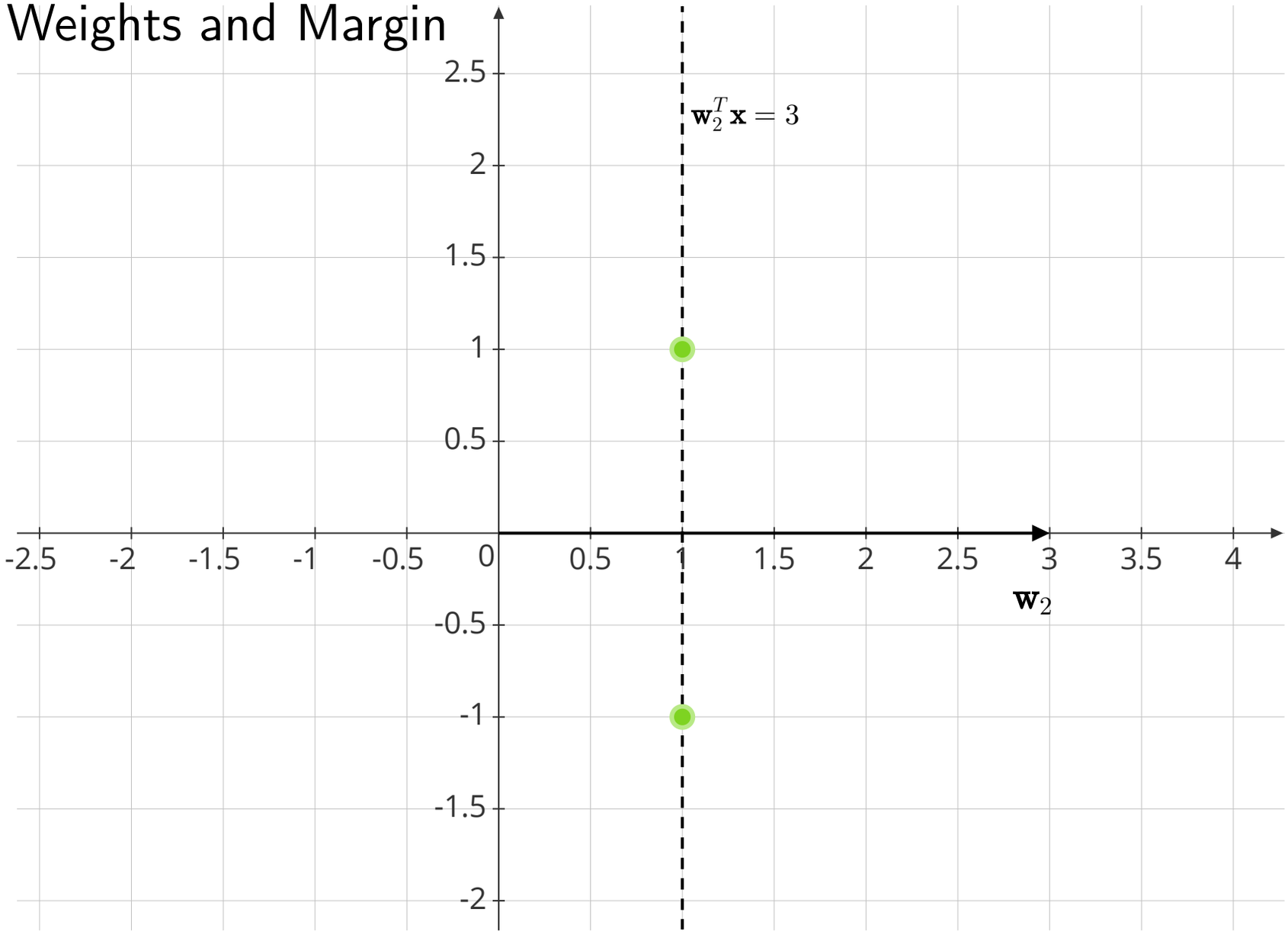
$$\mathbf{w}_2 = \begin{bmatrix} 3 & 0 \end{bmatrix}^T$$

Weights and Margin



$$\mathbf{w}_2 = \begin{bmatrix} 3 & 0 \end{bmatrix}^T$$

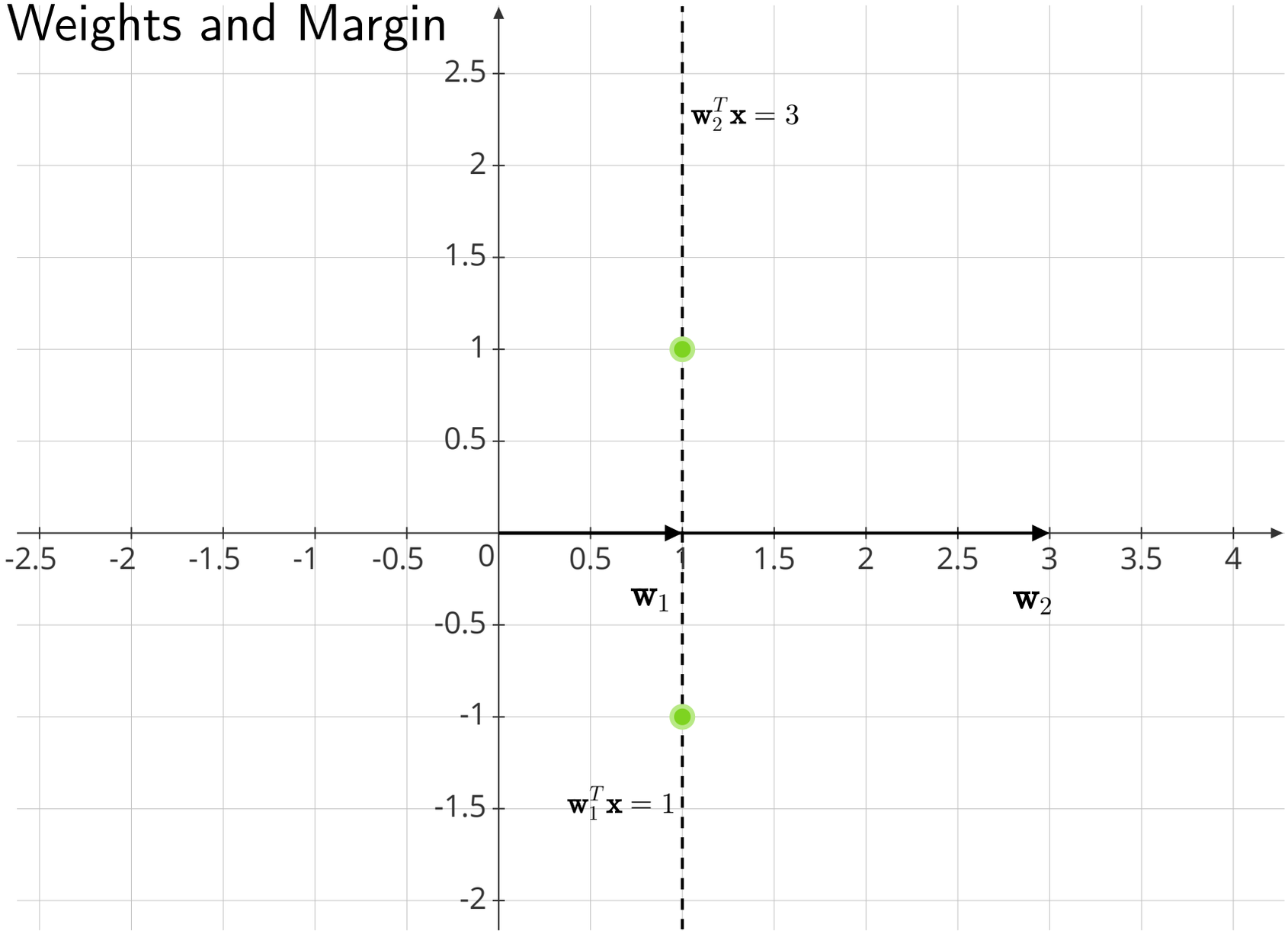
Weights and Margin



$$\mathbf{w}_2 = \begin{bmatrix} 3 & 0 \end{bmatrix}^T$$

$$\mathbf{w}_2^T\mathbf{x} = 3 \Longrightarrow \frac{1}{3}\left(\mathbf{w}_2^T\mathbf{x}\right) = \frac{1}{3} \cdot 3$$

Weights and Margin



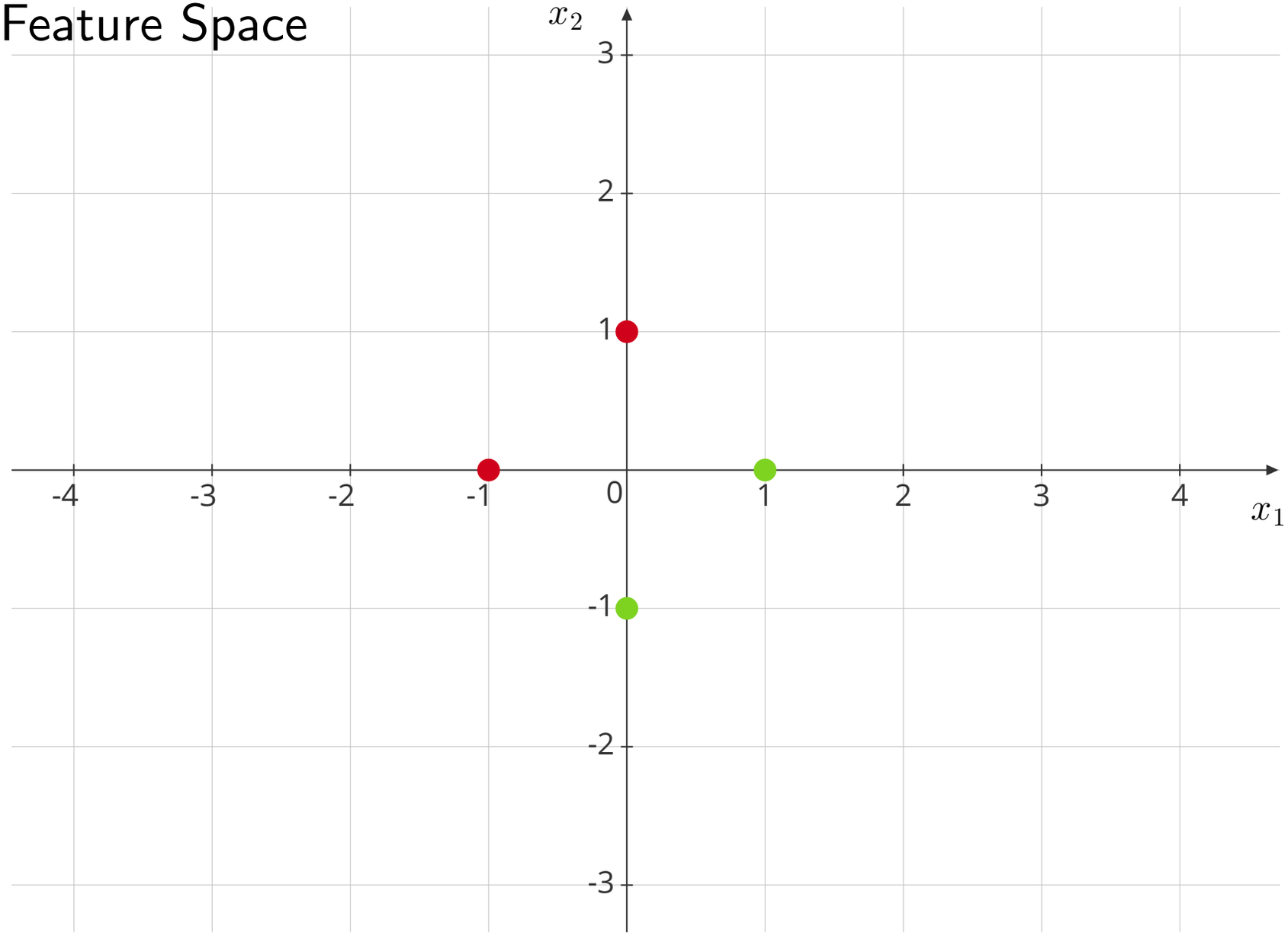
$$\mathbf{w}_2 = \begin{bmatrix} 3 & 0 \end{bmatrix}^T$$

$$\mathbf{w}_2^T\mathbf{x} = 3 \Longrightarrow \frac{1}{3}\left(\mathbf{w}_2^T\mathbf{x}\right) = \frac{1}{3}\cdot 3$$

$$\Longrightarrow \left(\frac{\mathbf{w}_2}{3}\right)^T \mathbf{x} = 1 \Longrightarrow \mathbf{w}_1^T\mathbf{x} = 1$$

$$\frac{\mathbf{w}_2}{3} = \begin{bmatrix} 1 & 0 \end{bmatrix}^T = \mathbf{w}_1$$

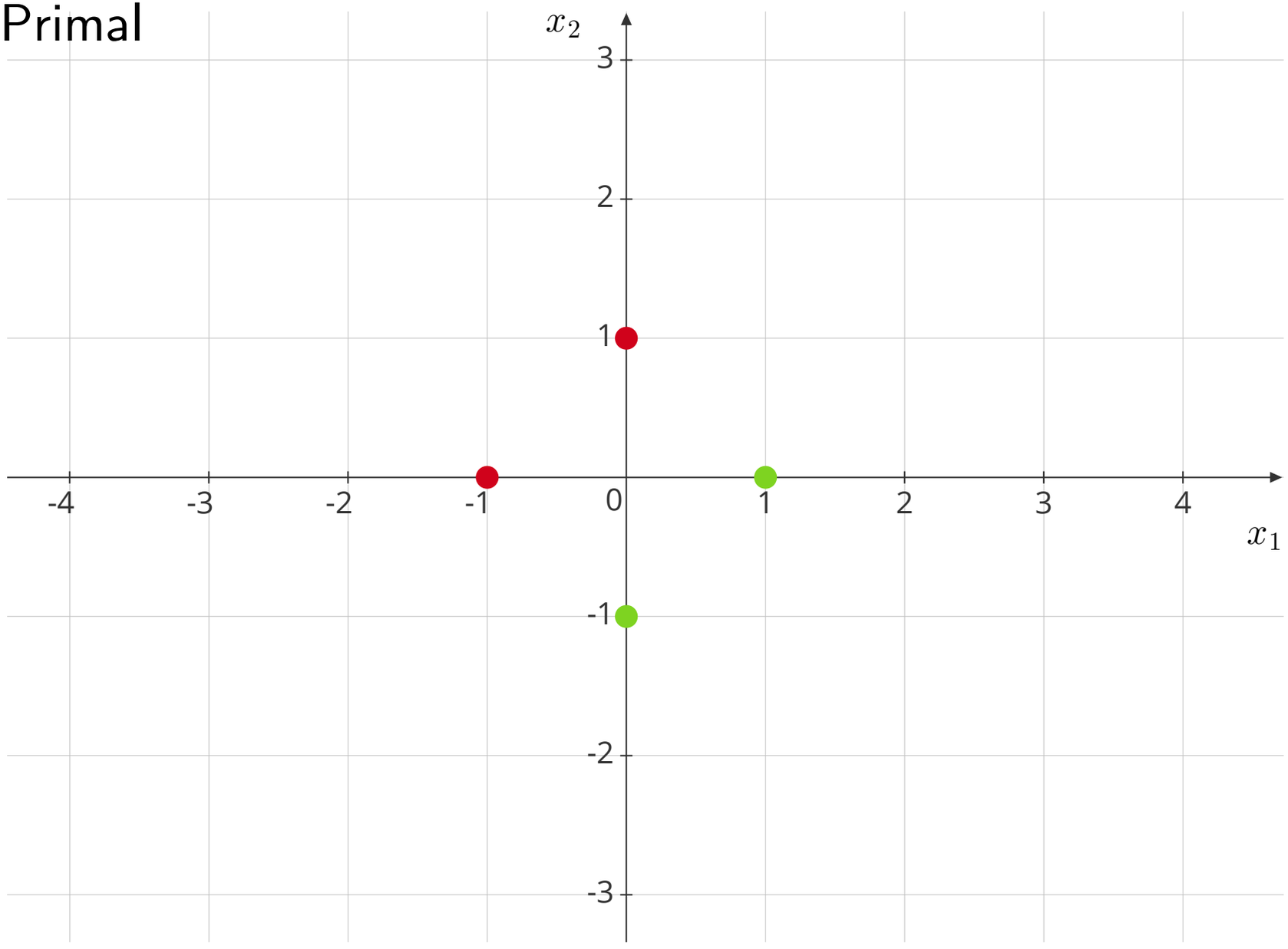
Feature Space



Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

Primal



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$$\min_{\mathbf{w}} \quad \frac{||\mathbf{w}||^2}{2}$$

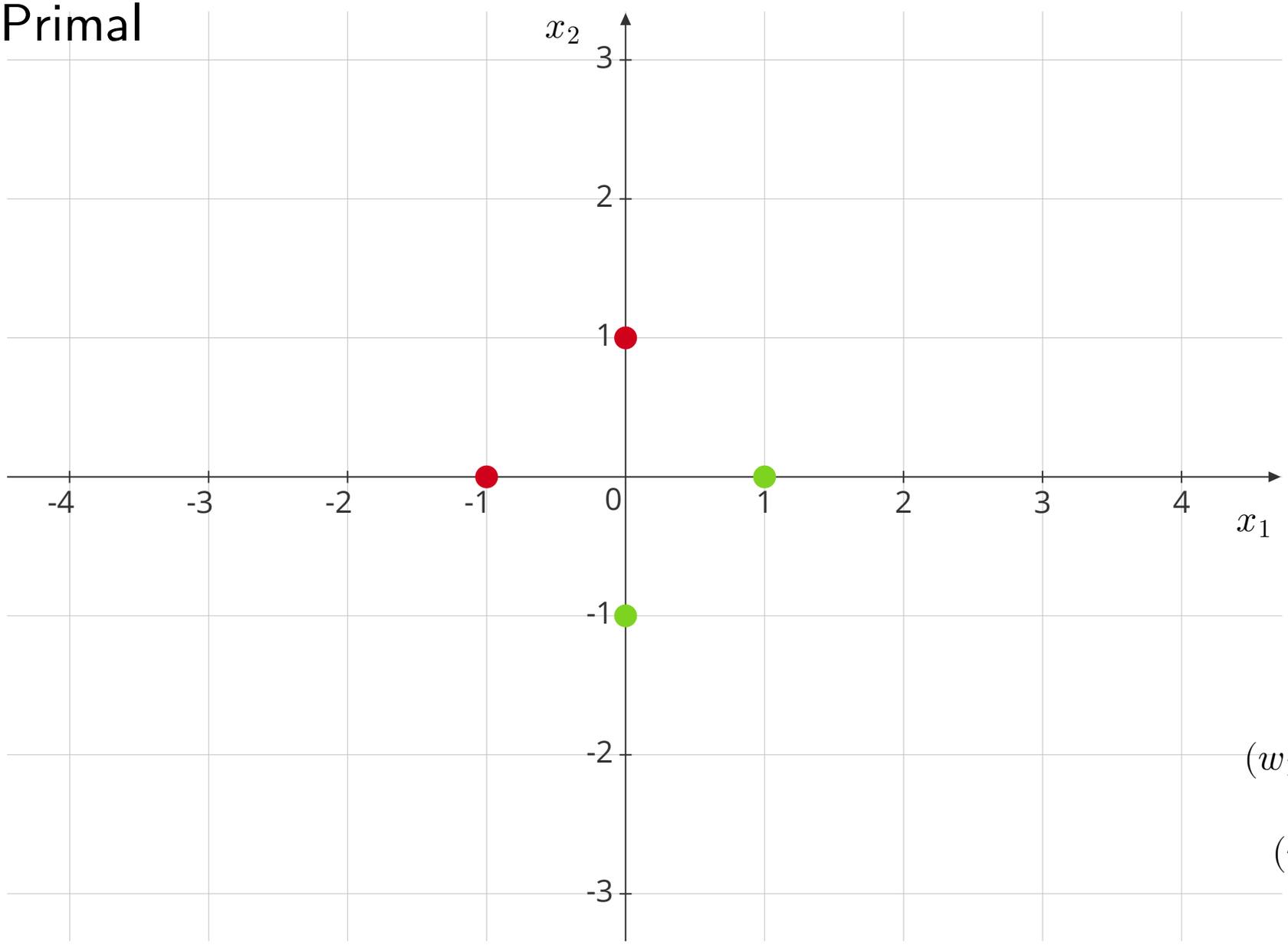
$$\left(\mathbf{w}^T \mathbf{x}_1\right) y_1 \geqslant 1 \quad (1)$$

$$\left(\mathbf{w}^T \mathbf{x}_2\right) y_2 \geqslant 1 \quad (2)$$

$$\left(\mathbf{w}^T \mathbf{x}_3\right) y_3 \geqslant 1 \quad (3)$$

$$\left(\mathbf{w}^T \mathbf{x}_4\right) y_4 \geqslant 1 \quad (4)$$

Primal



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$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

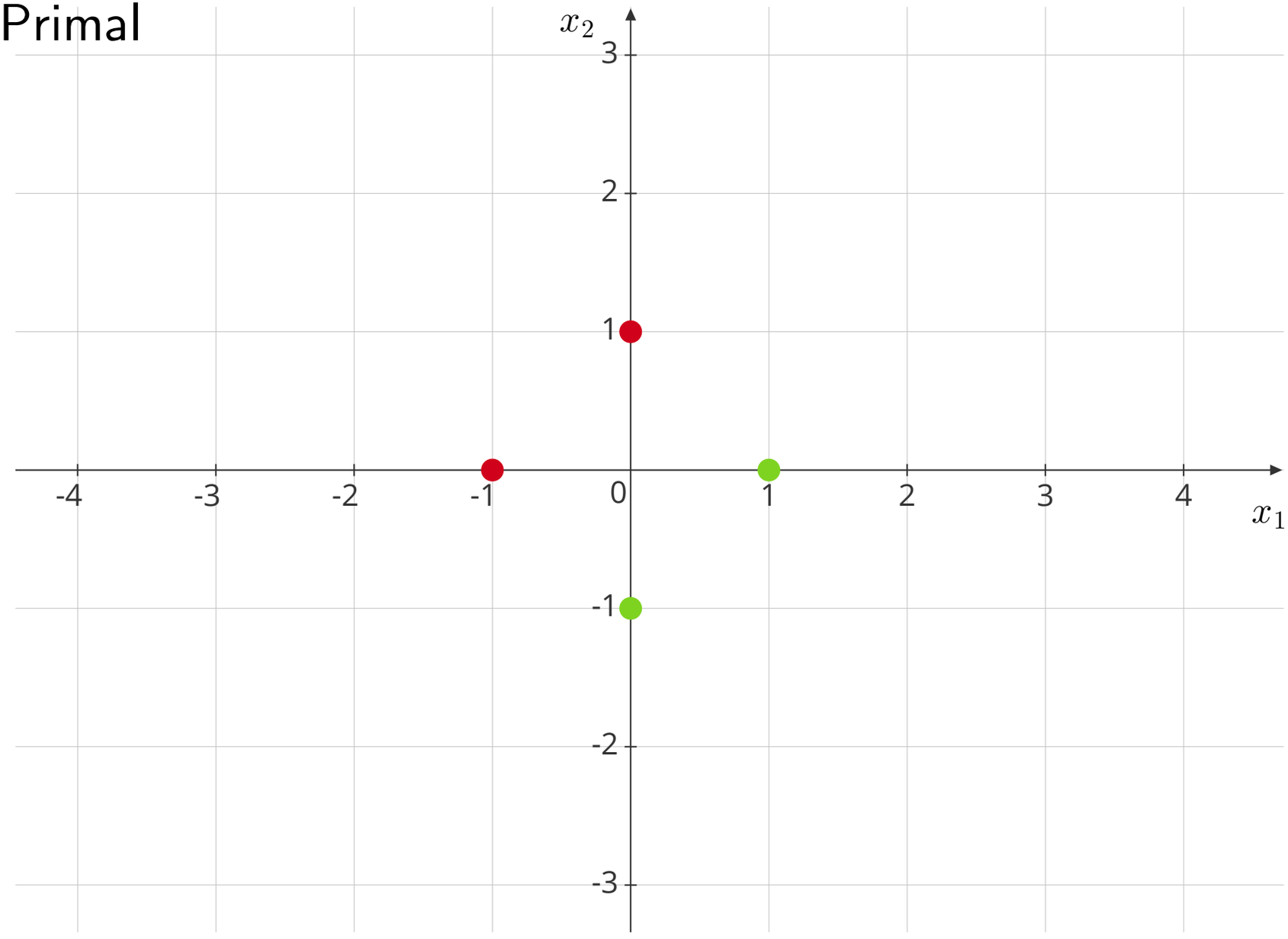
$$(w_1 \times 1 + w_2 \times 0) \cdot 1 \geqslant 1 \quad (1)$$

$$(w_1 \times 0 + w_2 \times -1) \cdot 1 \geqslant 1 \quad (2)$$

$$(w_1 \times (-1) + w_2 \times 0) \cdot (-1) \geqslant 1 \quad (3)$$

$$(w_1 \times (0) + w_2 \times 1) \cdot (-1) \geqslant 1 \quad (4)$$

Primal



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$$w_1 \geqslant 1 \qquad (1)$$

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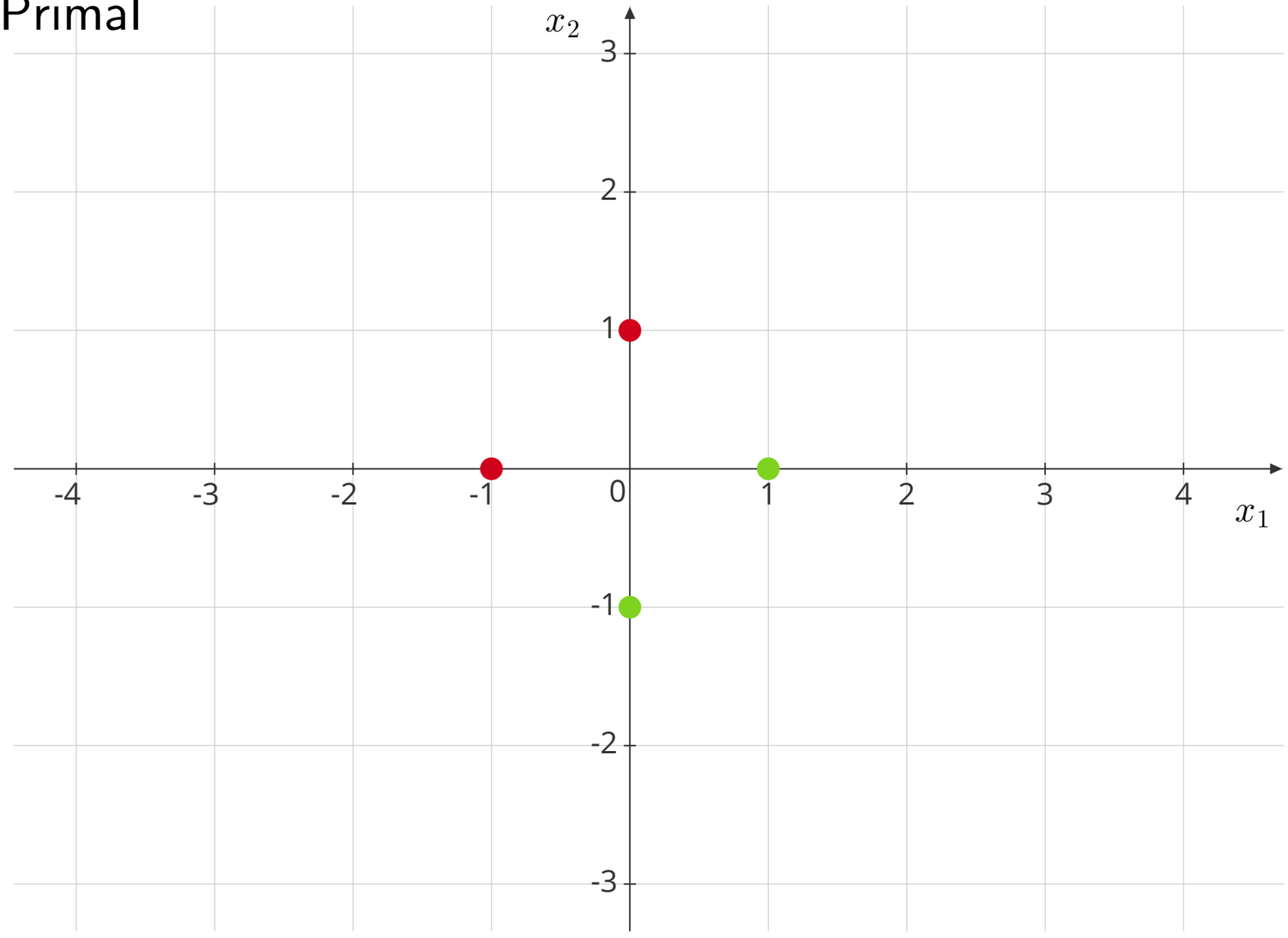

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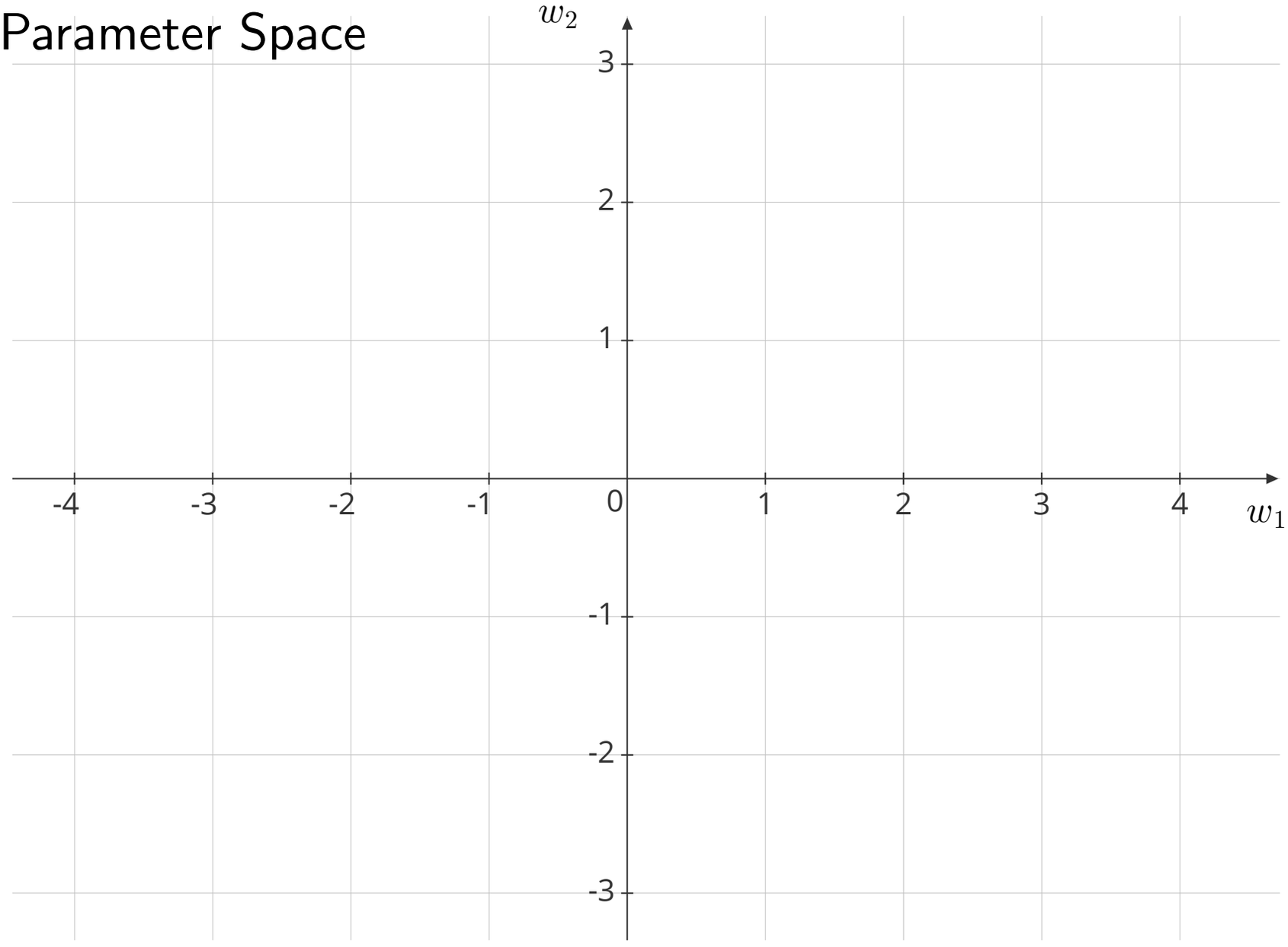
Parameter Space



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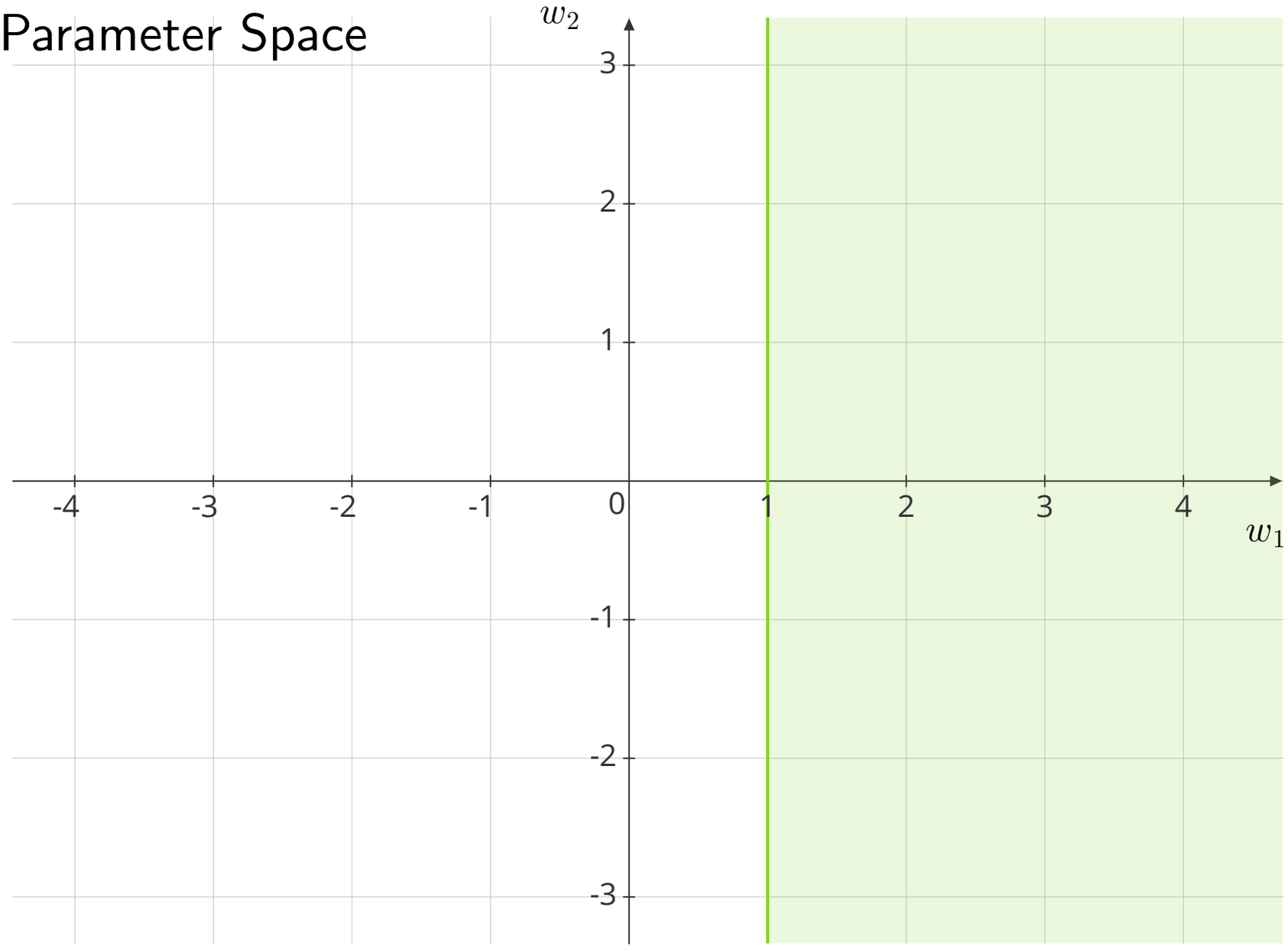

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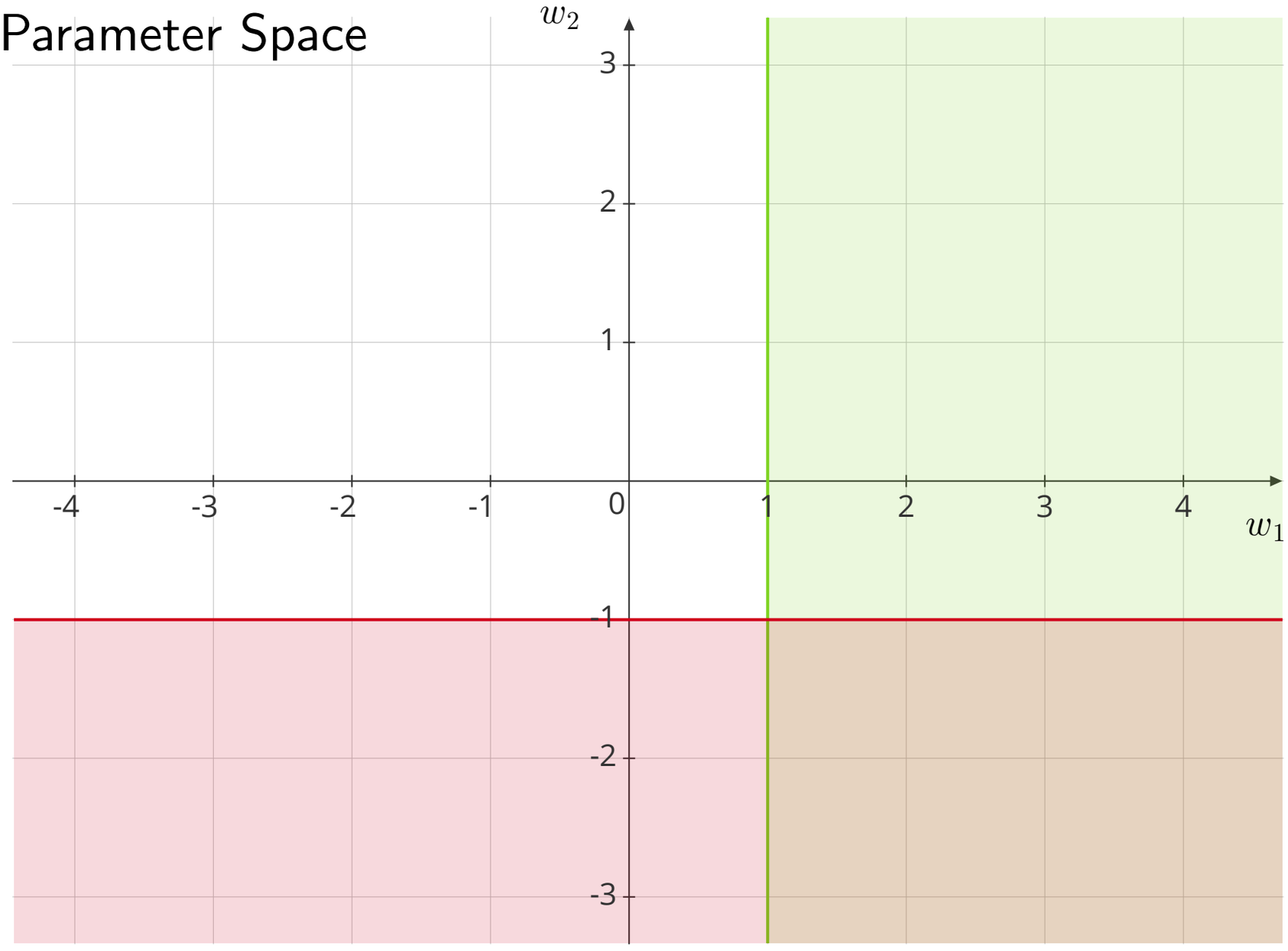

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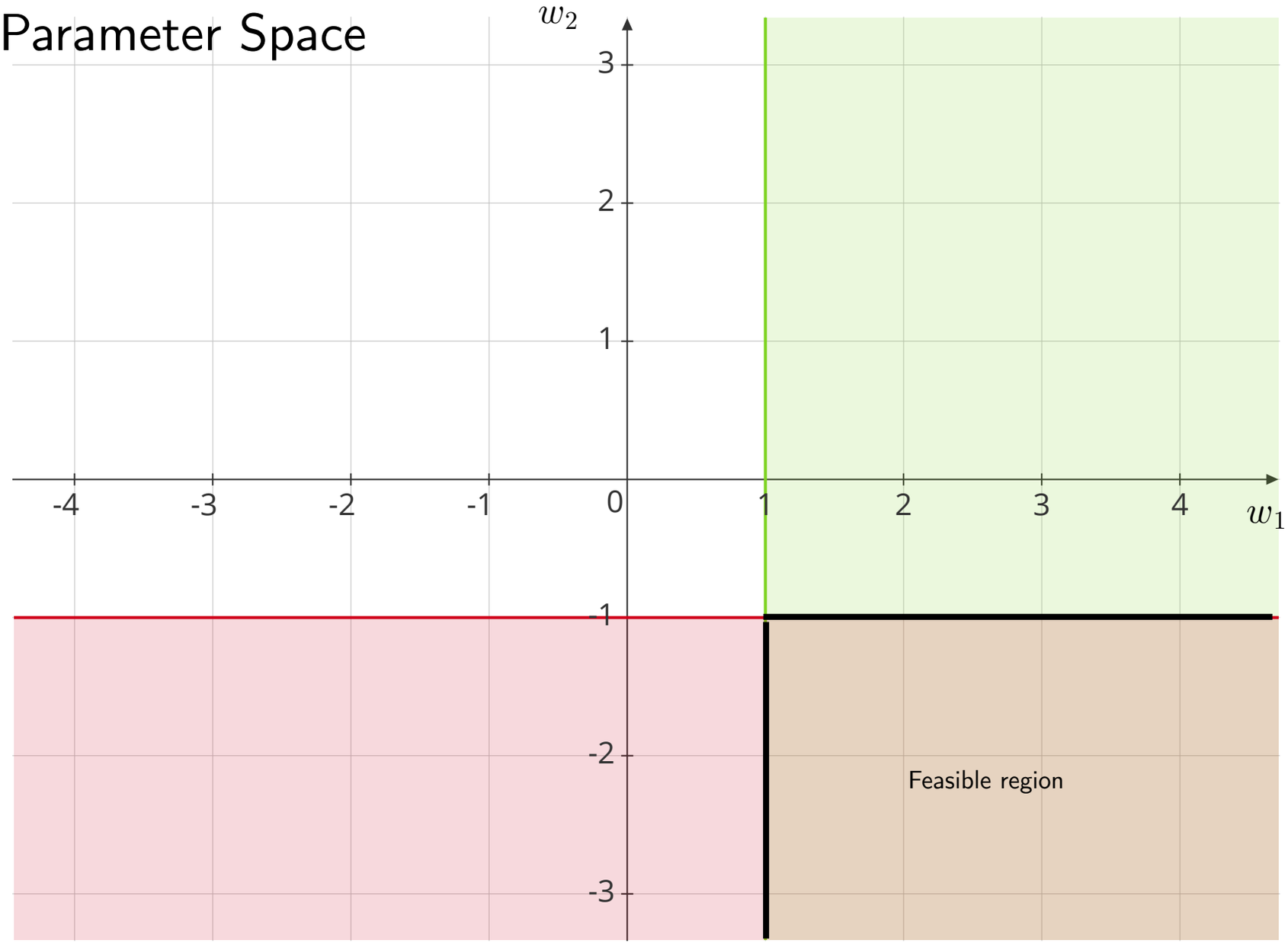

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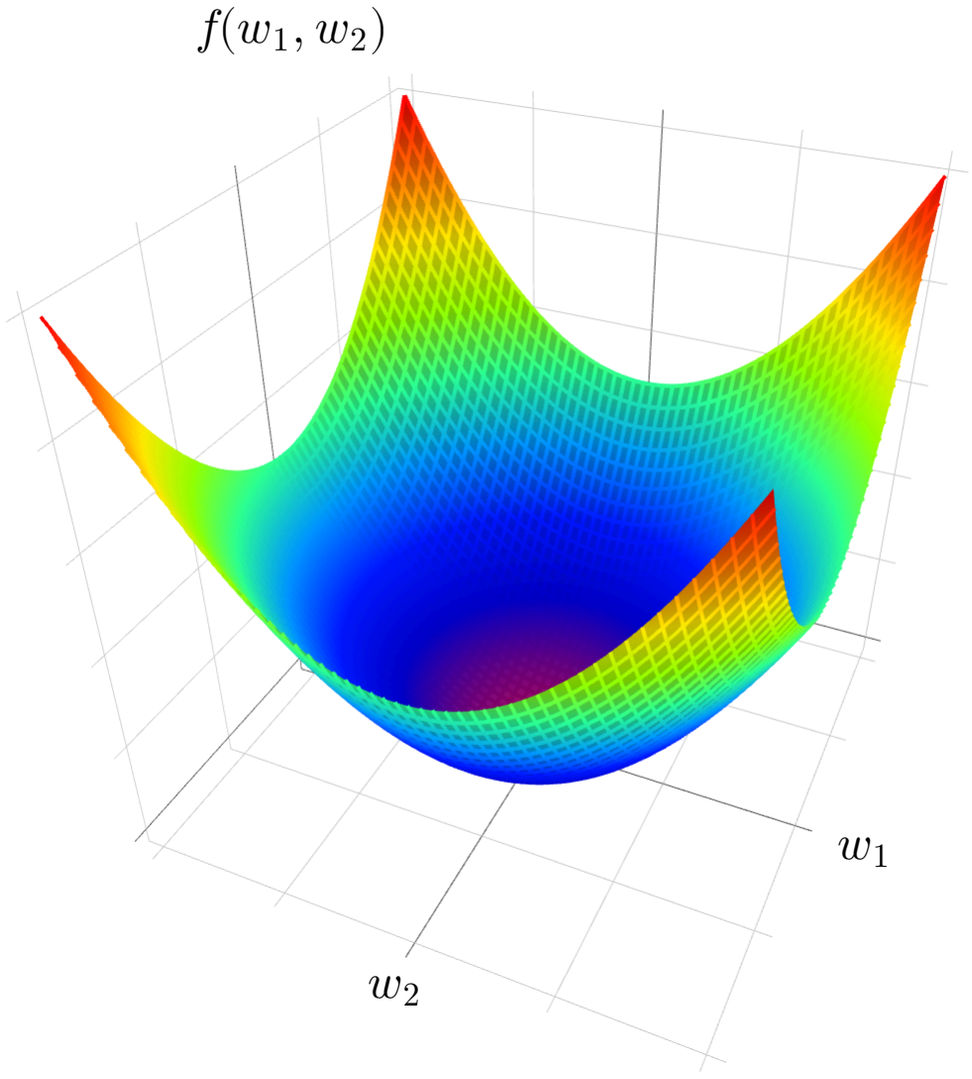
Parameter Space - Objective Surface

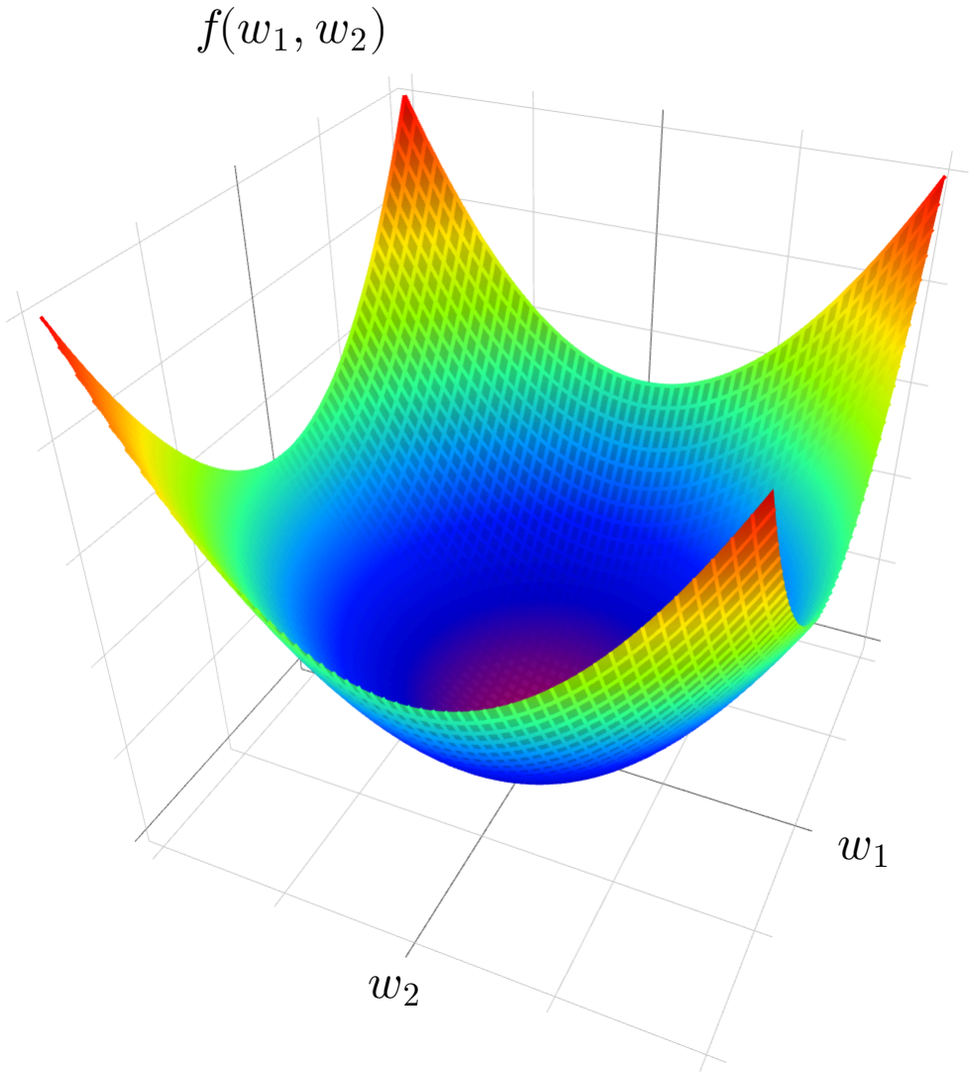
Example-1



$$f(w_1, w_2) = \frac{||\mathbf{w}||^2}{2} = \frac{w_1^2 + w_2^2}{2}$$

Parameter Space - Objective Surface

Example-1



$$f(w_1, w_2) = \frac{||\mathbf{w}||^2}{2} = \frac{w_1^2 + w_2^2}{2}$$

$$f(w_1, w_2) = 0.5 \implies w_1^2 + w_2^2 = 1$$

Parameter Space - Objective Surface

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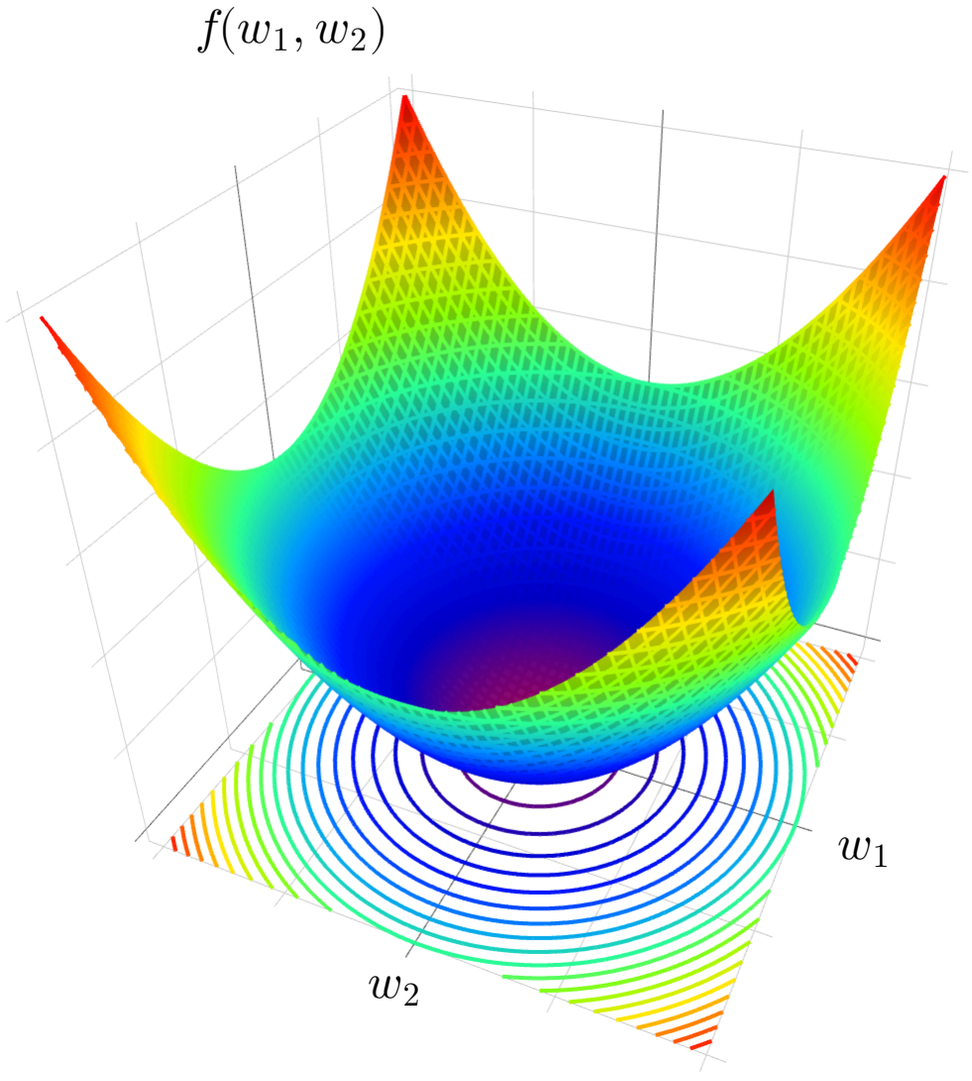

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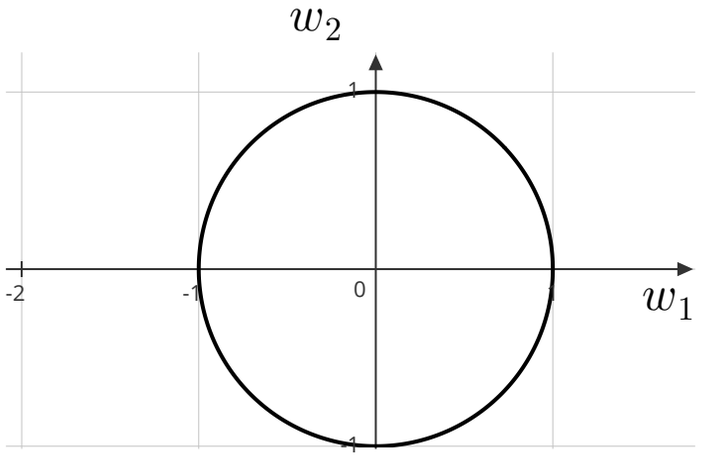

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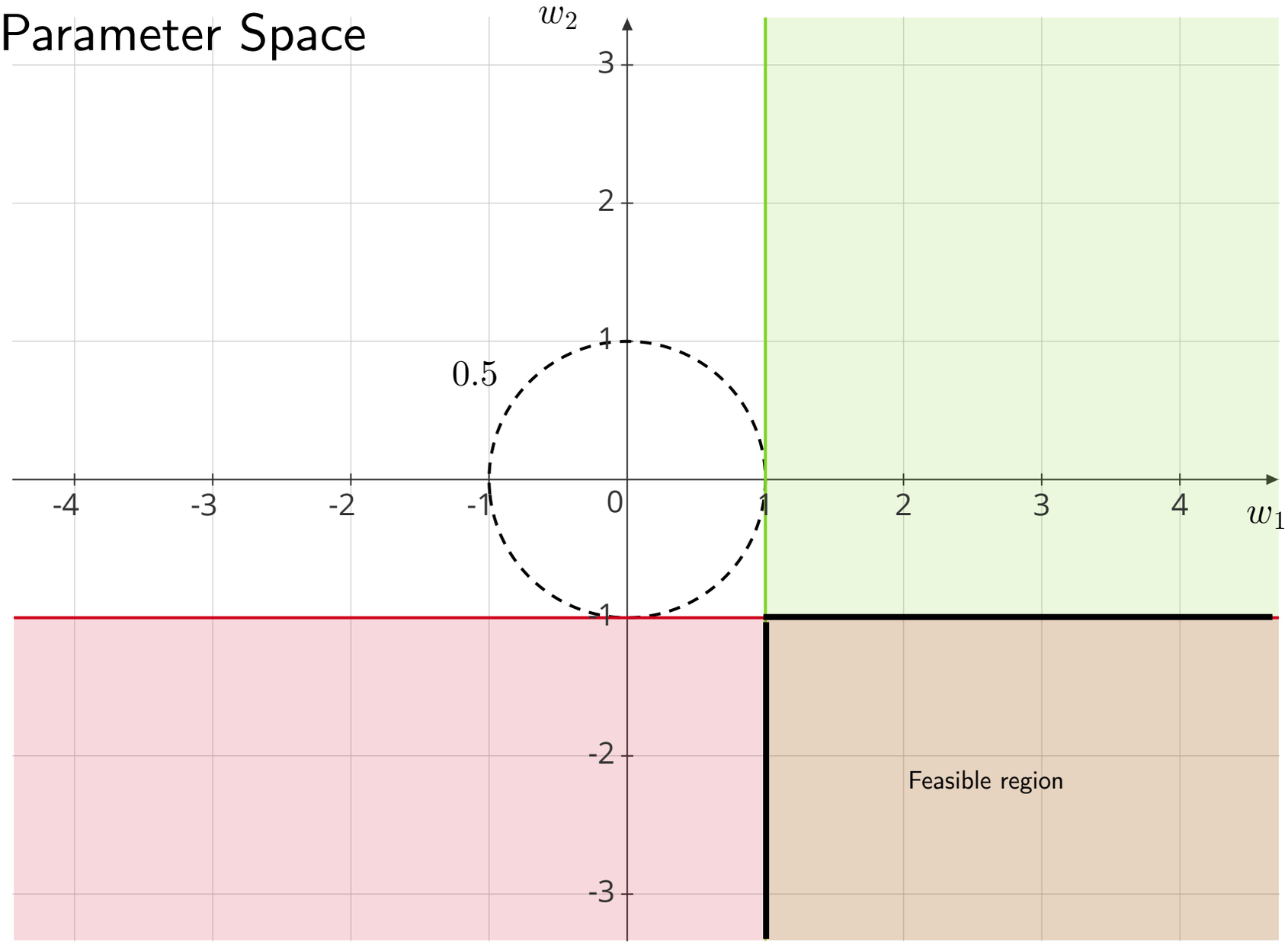
Parameter Space



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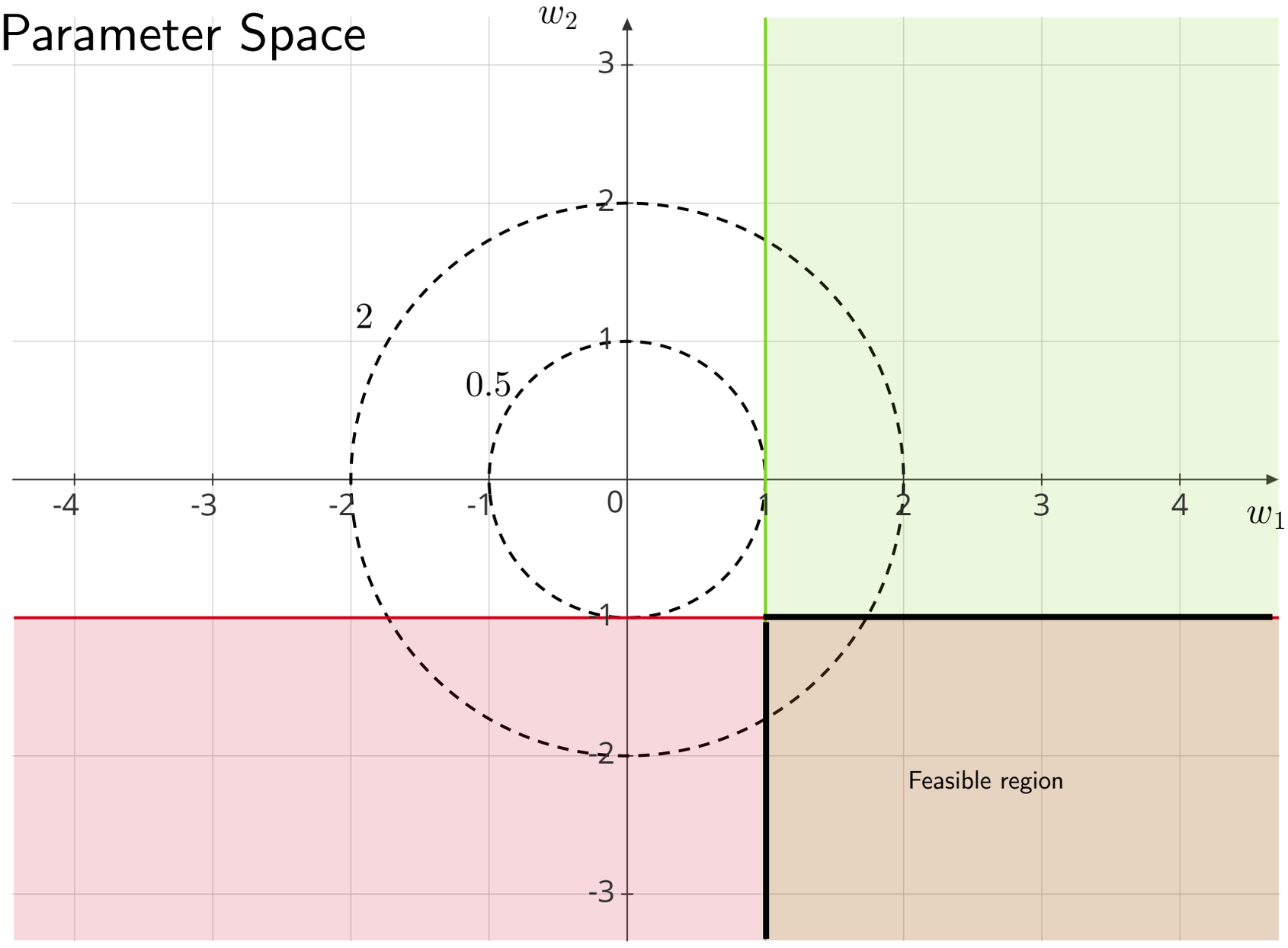

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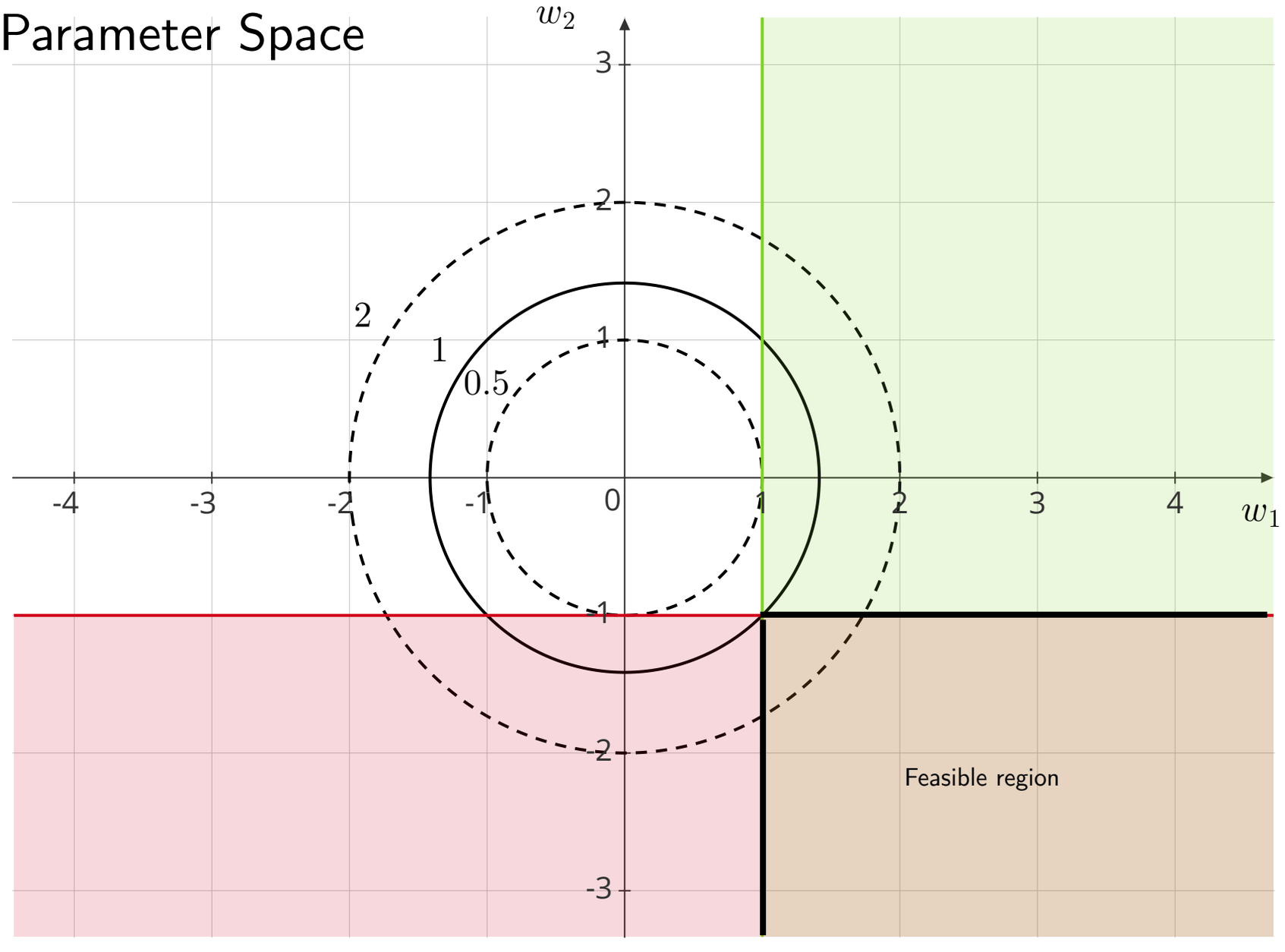

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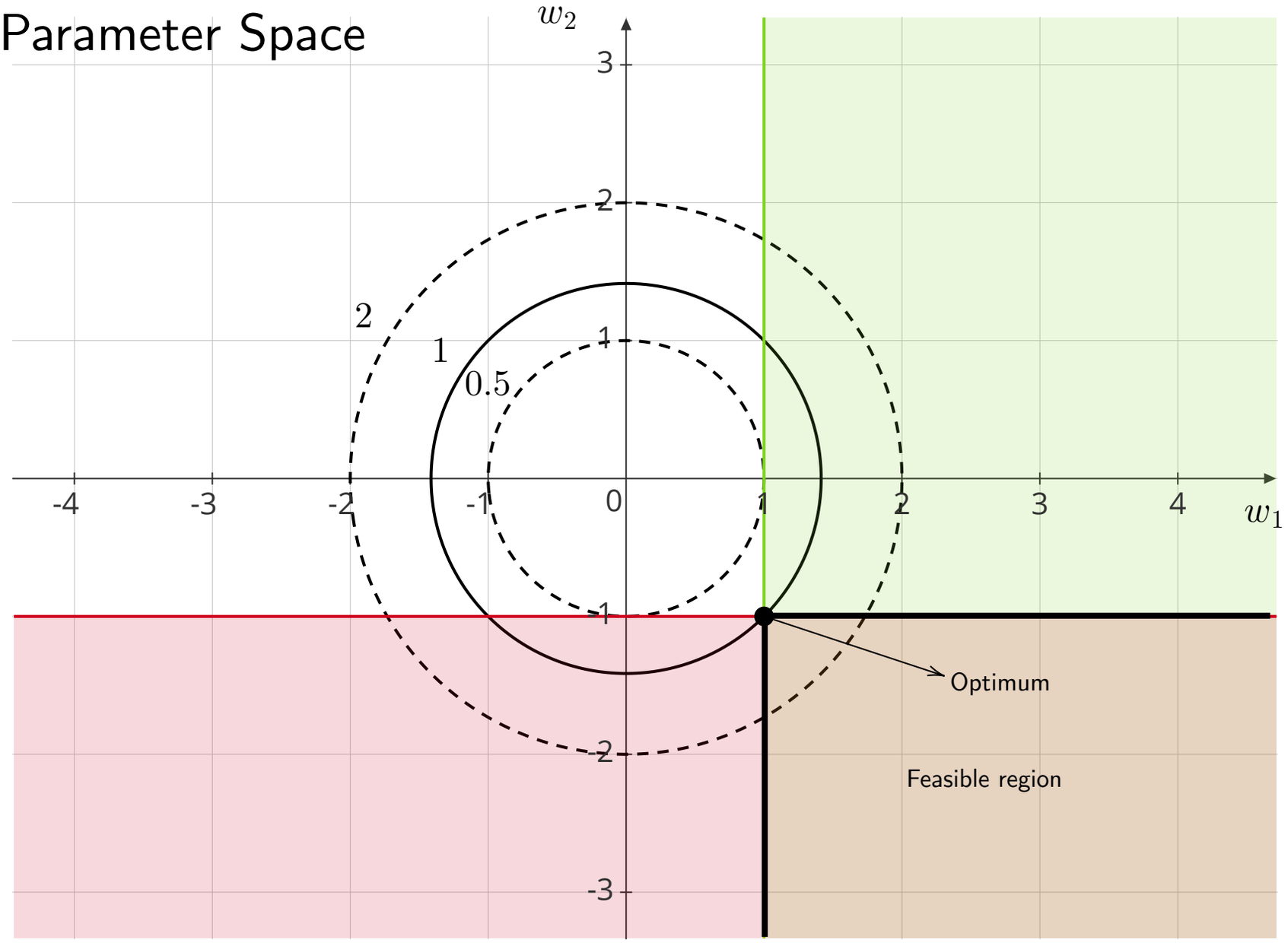

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Primal $\rightarrow$ Dual

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$$\max_{\boldsymbol{\alpha} \geqslant 0} \; \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \cdot \boldsymbol{\alpha}^T \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y} \boldsymbol{\alpha}$$

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}$$

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$$\mathbf{Y} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$\mathbf{Q} = \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y}$$

$$\mathbf{Q} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; (\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4) - \frac{1}{2}\left(\alpha_1^2 + \alpha_2^2 + \alpha_3^2 + \alpha_4^2\right) - (\alpha_1 \alpha_3 + \alpha_2 \alpha_4)$$

Primal $\rightarrow$ Dual

Example-1

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \cdot \boldsymbol{\alpha}^T \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y} \boldsymbol{\alpha}$$

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$\mathbf{Q} = \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y}$$

$$\mathbf{Q} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; (\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4) - \frac{1}{2}\left(\alpha_1^2 + \alpha_2^2 + \alpha_3^2 + \alpha_4^2\right) - (\alpha_1 \alpha_3 + \alpha_2 \alpha_4)$$

$$\boldsymbol{\alpha}^* = \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \end{bmatrix}^T$$

Primal $\rightarrow$ Dual

Example-1

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \cdot \boldsymbol{\alpha}^T \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y} \boldsymbol{\alpha}$$

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$\mathbf{Q} = \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y}$$

$$\mathbf{Q} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

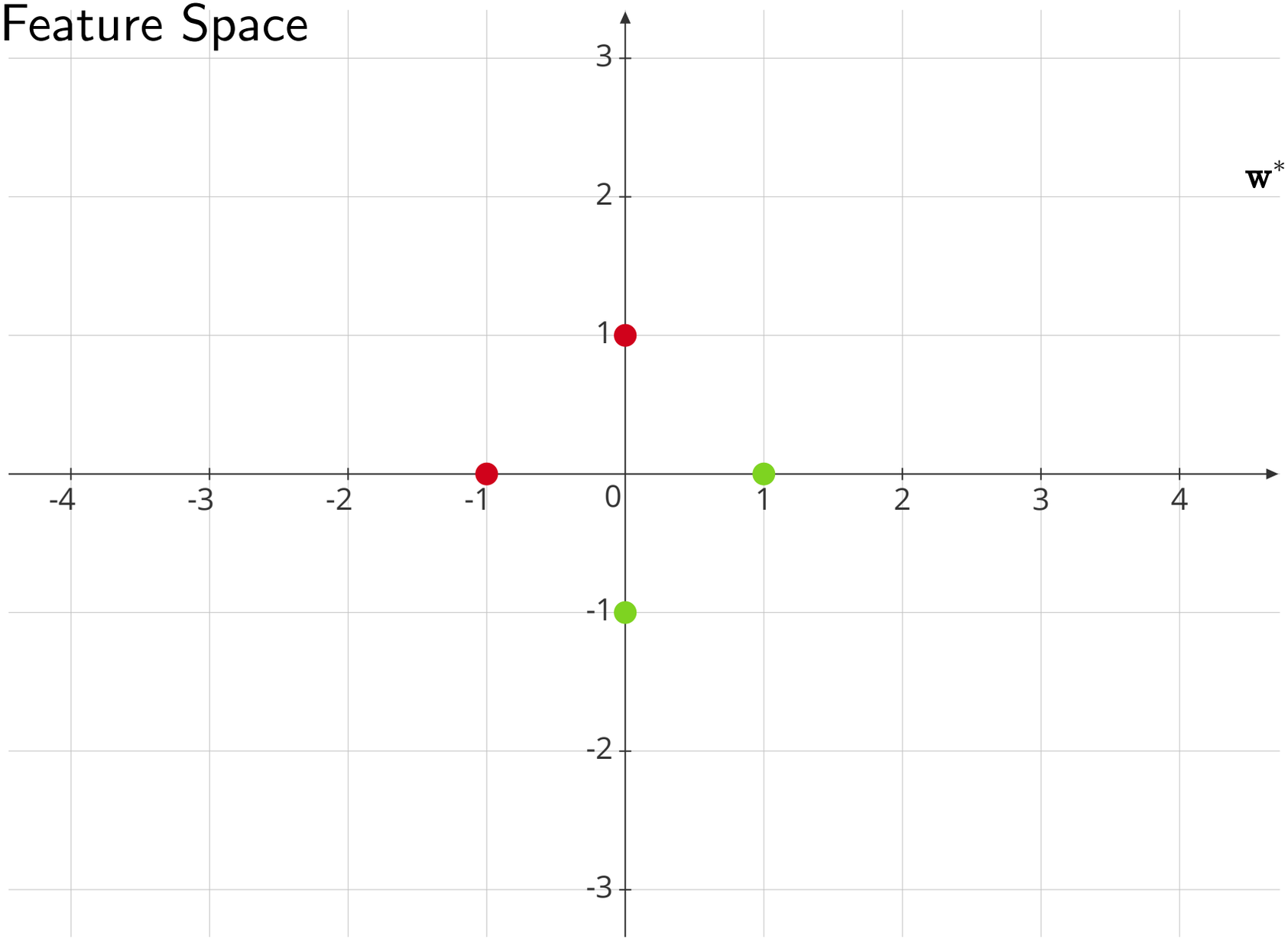
$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; (\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4) - \frac{1}{2}\left(\alpha_1^2 + \alpha_2^2 + \alpha_3^2 + \alpha_4^2\right) - (\alpha_1 \alpha_3 + \alpha_2 \alpha_4)$$

$$\boldsymbol{\alpha}^* = \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \end{bmatrix}^T \quad \text{OR} \quad \boldsymbol{\alpha}^* = \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix}^T$$

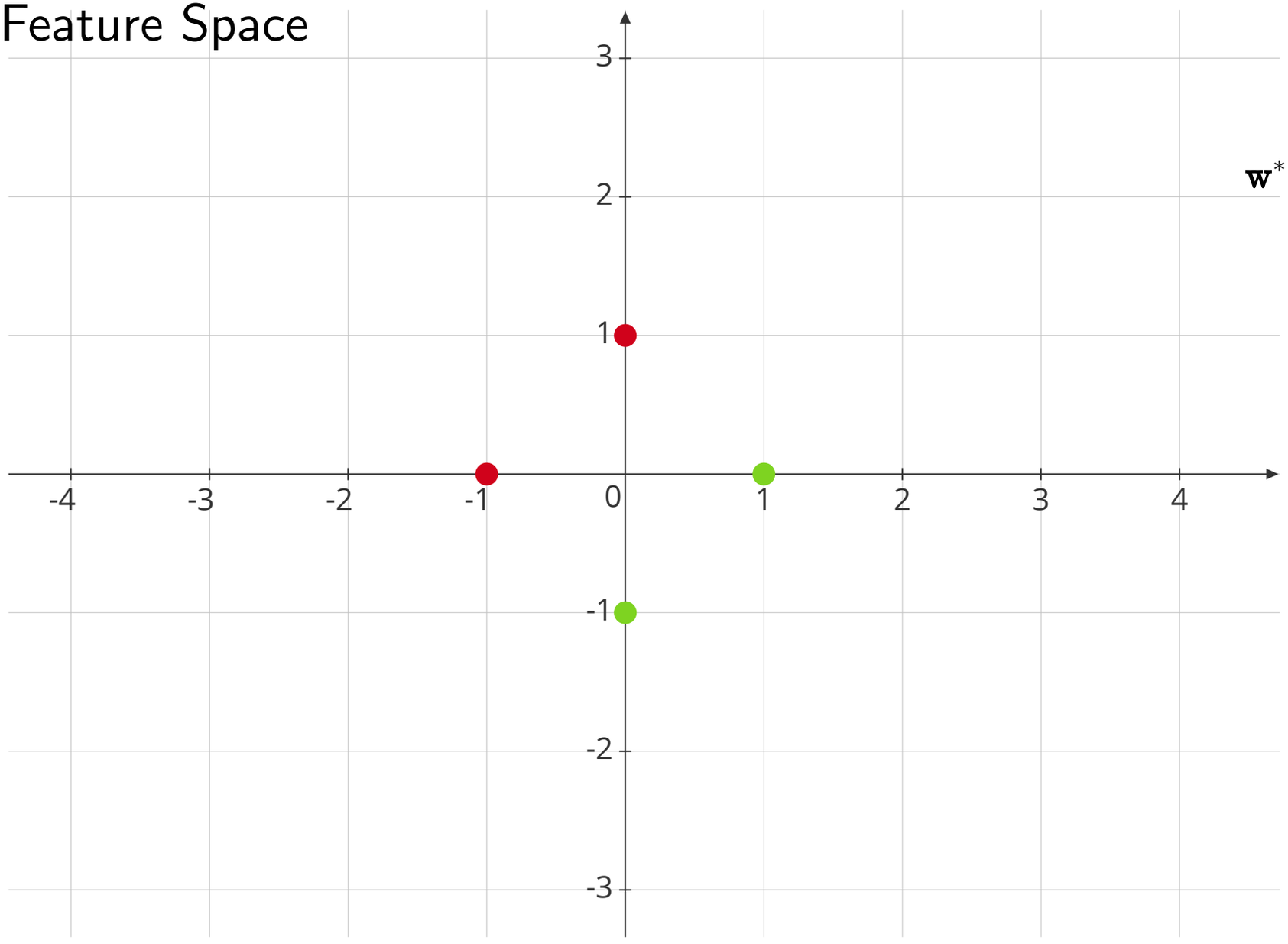
Feature Space

Example-1



$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

Feature Space

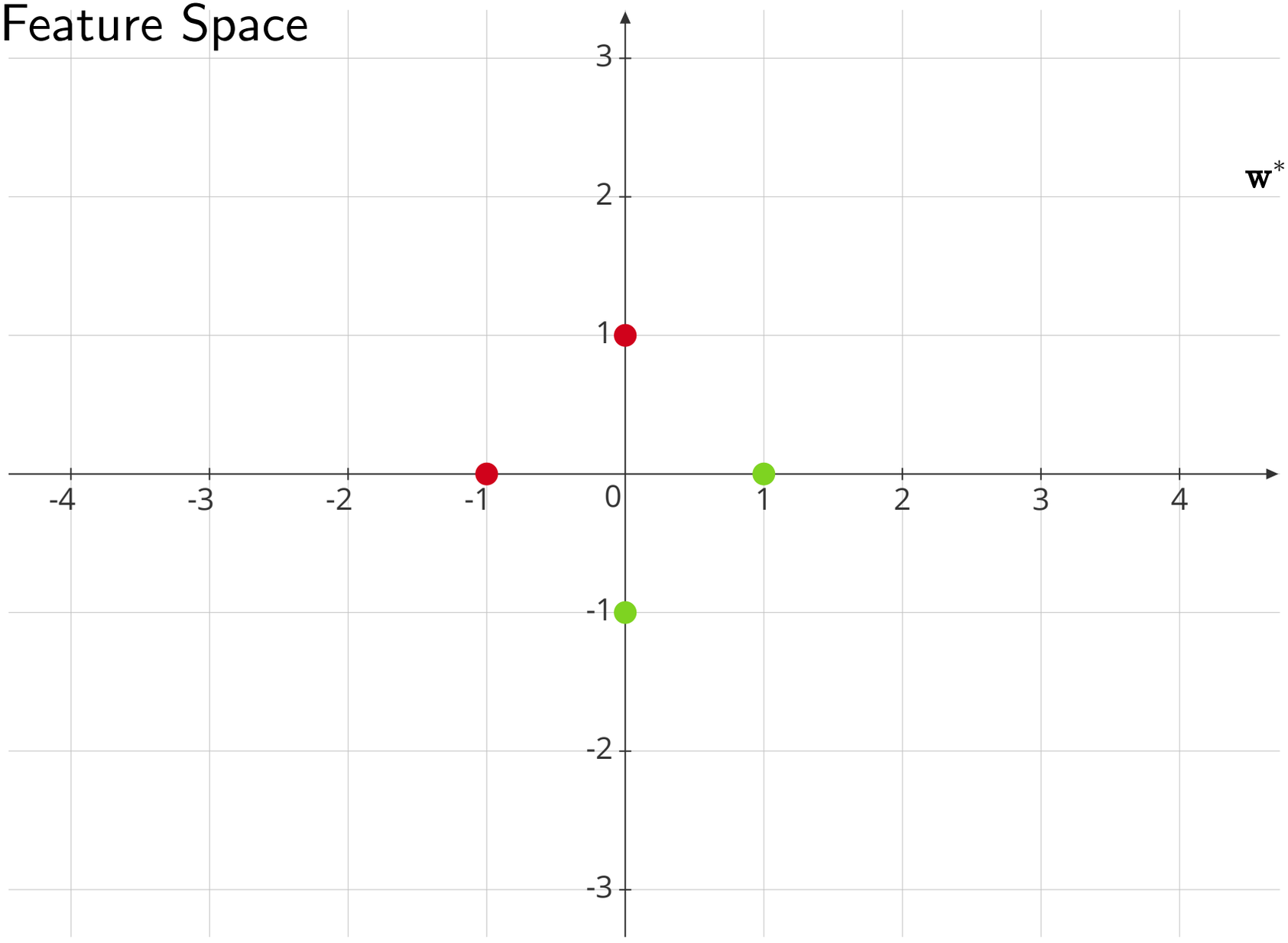


Example-1

$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

Feature Space



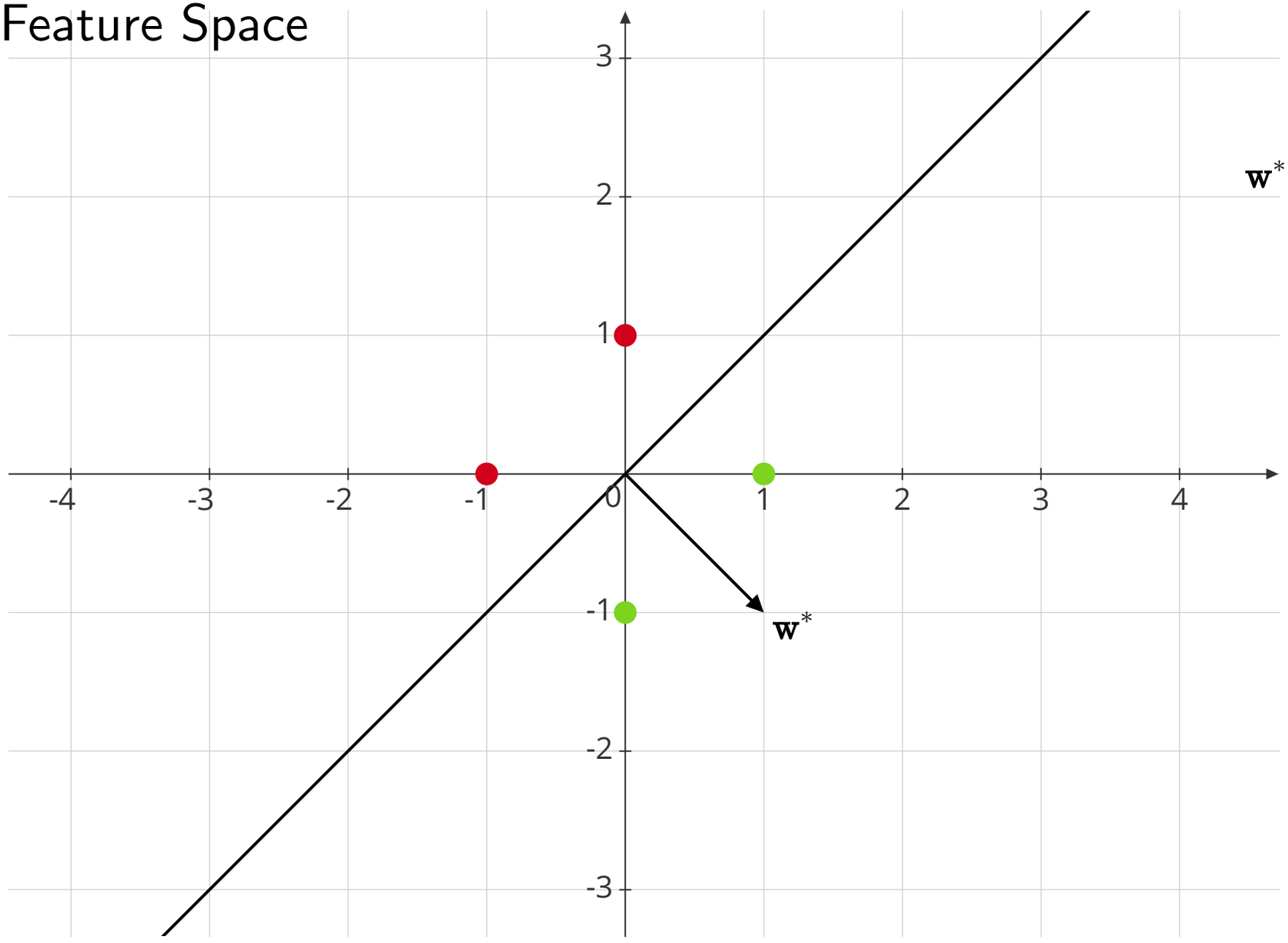
Example-1

$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space



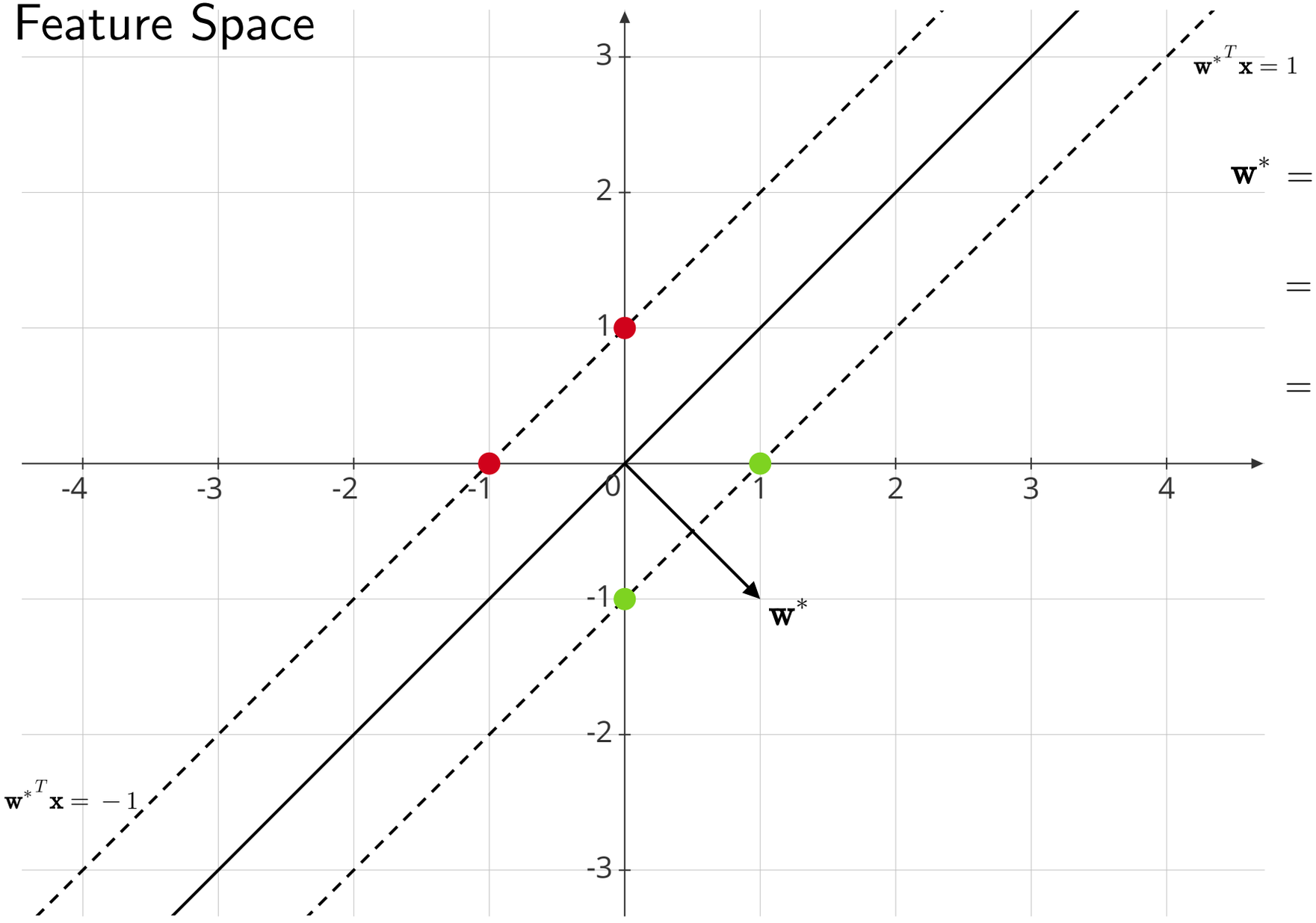
Example-1

$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space



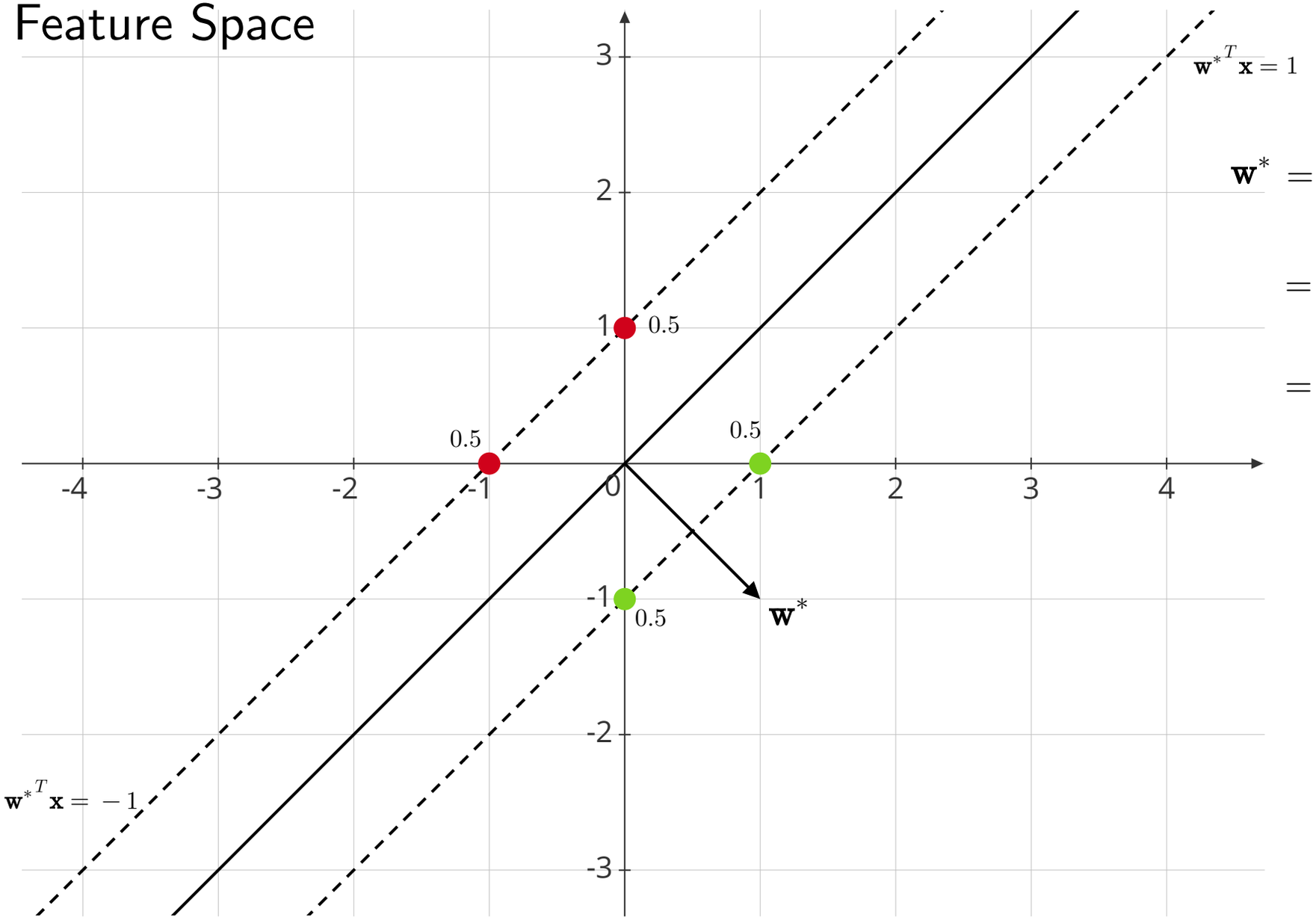
Example-1

$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space



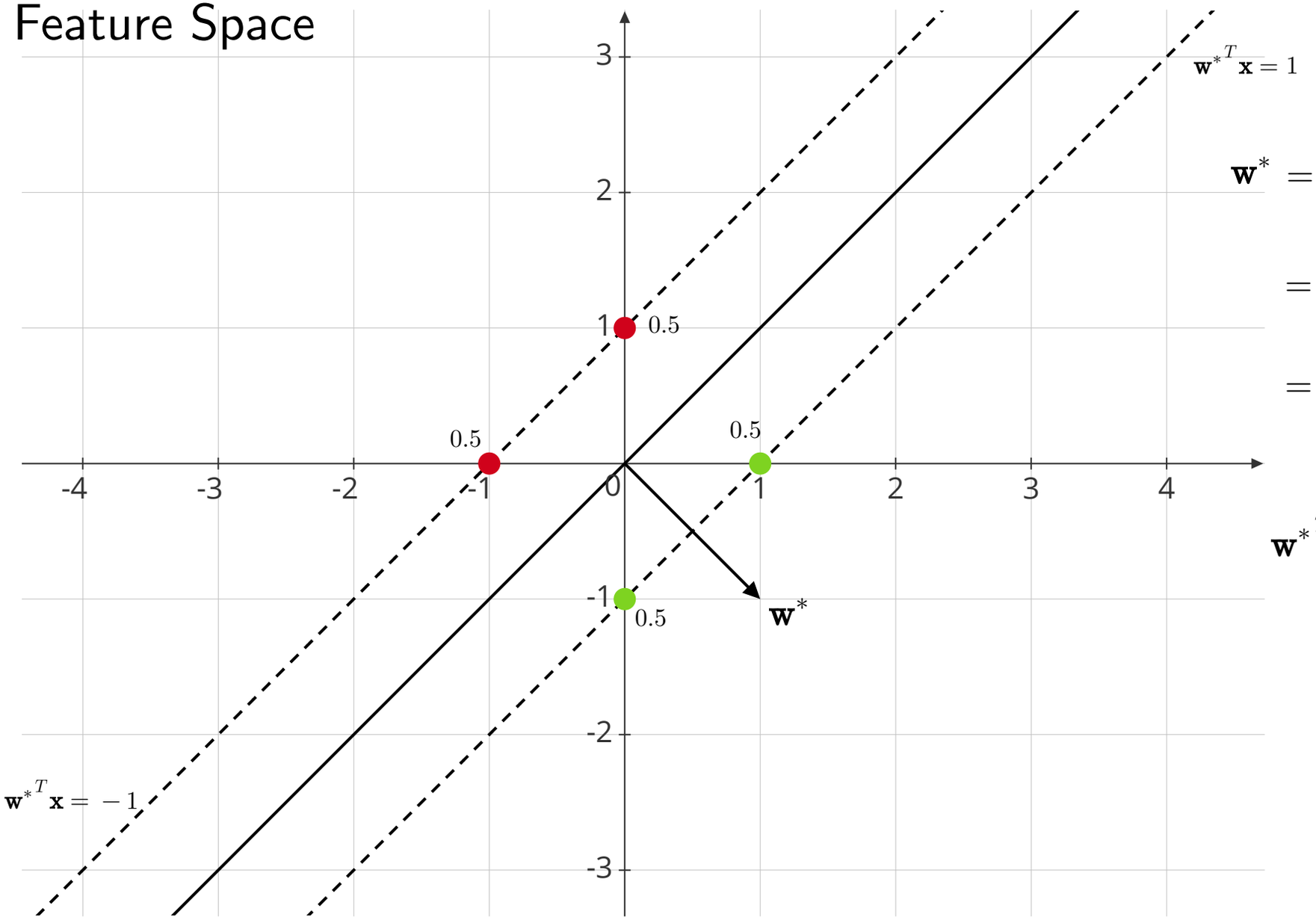
Example-1

$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space



Example-1

$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

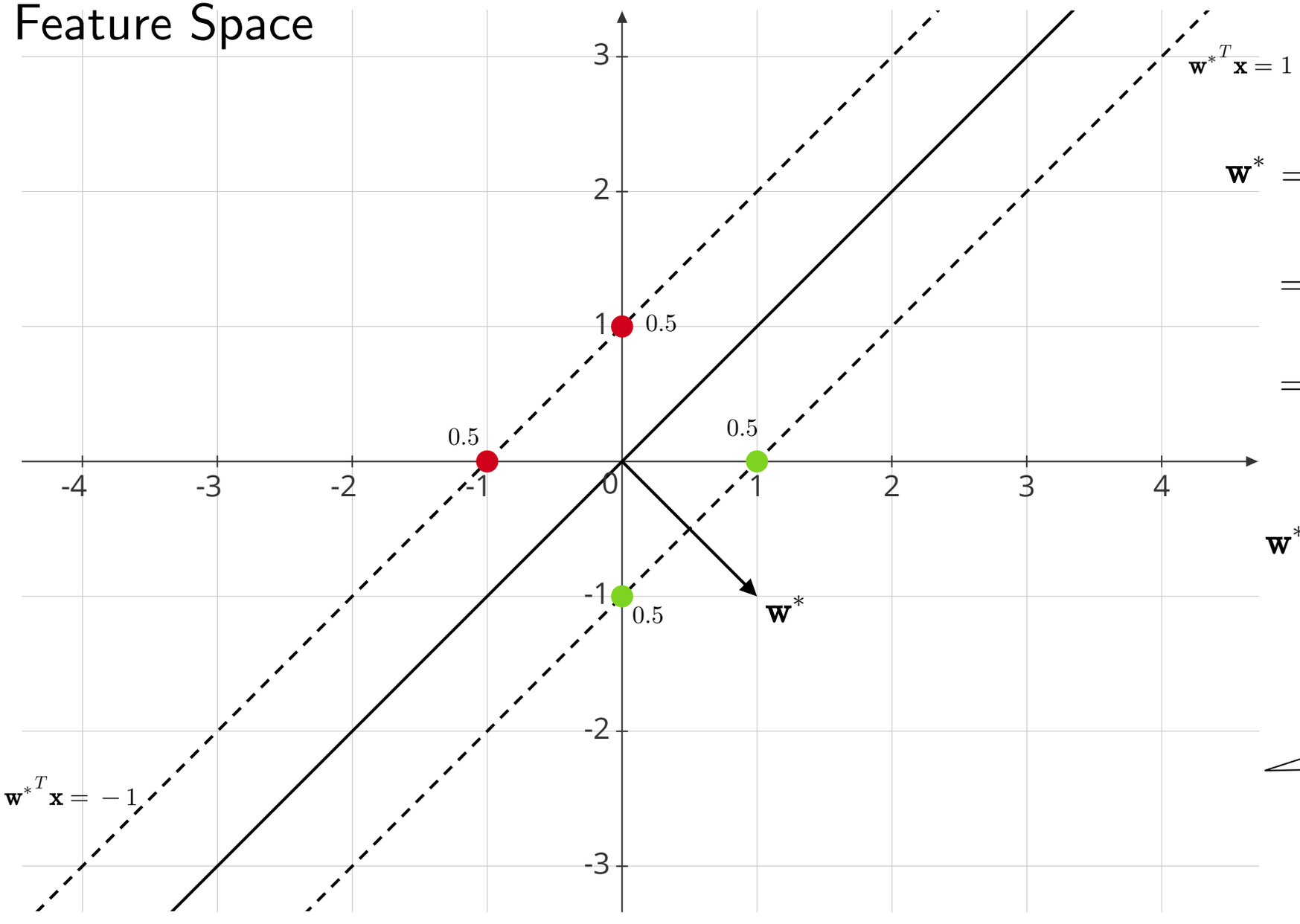
$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\mathbf{w}^{*T}\mathbf{x}_{\text{test}} = \sum_{i=1}^{n} \alpha_i \cdot \underbrace{\left[(y_i \mathbf{x}_i)^T \mathbf{x}_{\text{test}} \right]}_{\text{similarity}}$$

Feature Space

Example-1

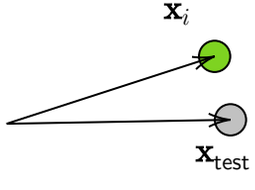


$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

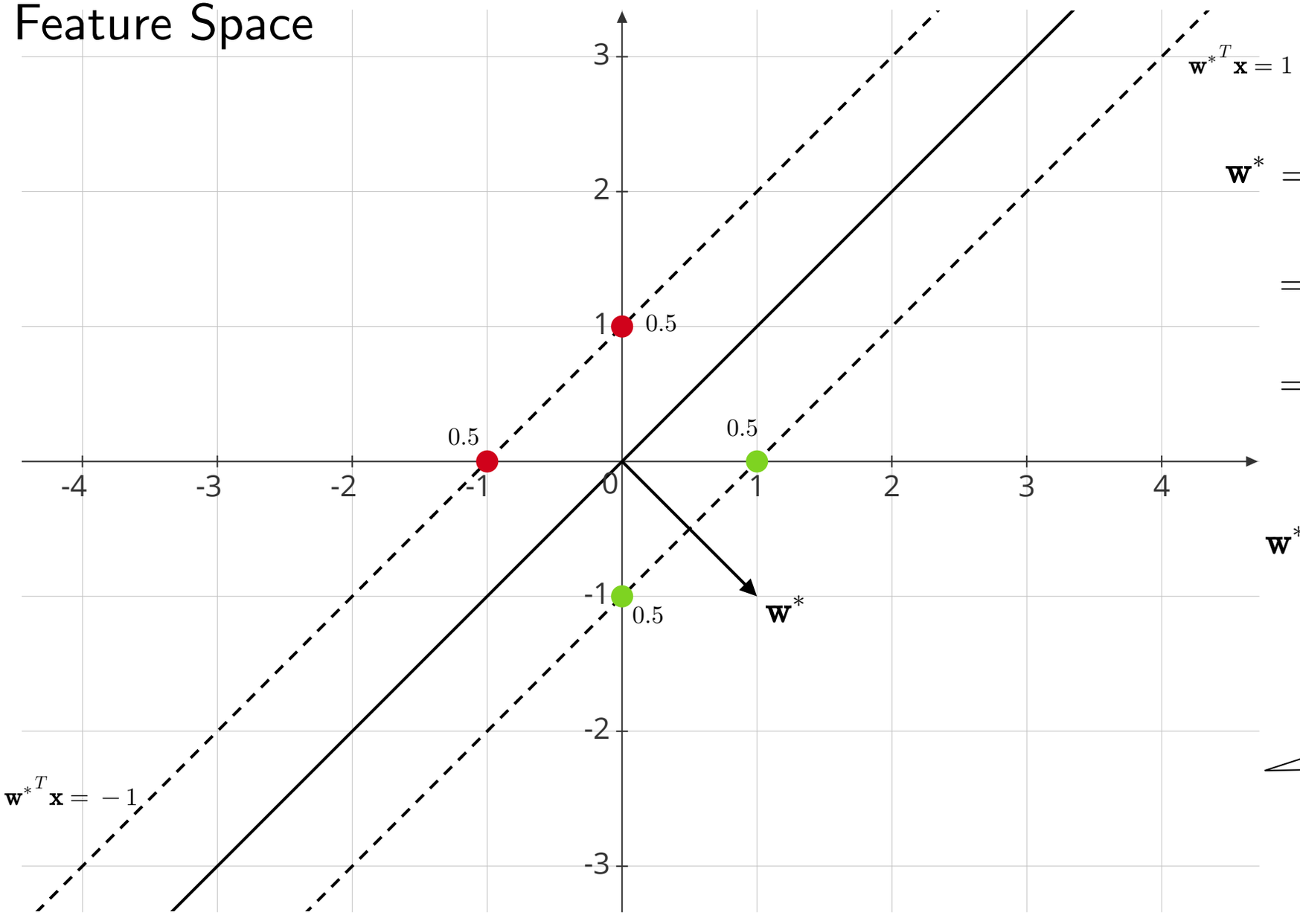
$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\mathbf{w}^{*T}\mathbf{x}_{\text{test}} = \sum_{i=1}^{n} \alpha_i \cdot \underbrace{\left[(y_i \mathbf{x}_i)^T \mathbf{x}_{\text{test}} \right]}_{\text{similarity}}$$



Feature Space

Example-1

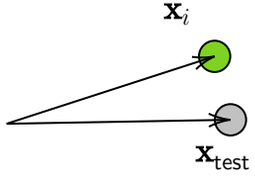


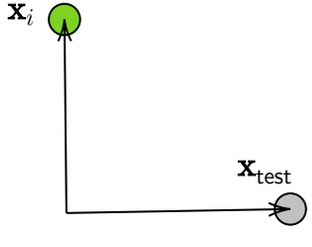
$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

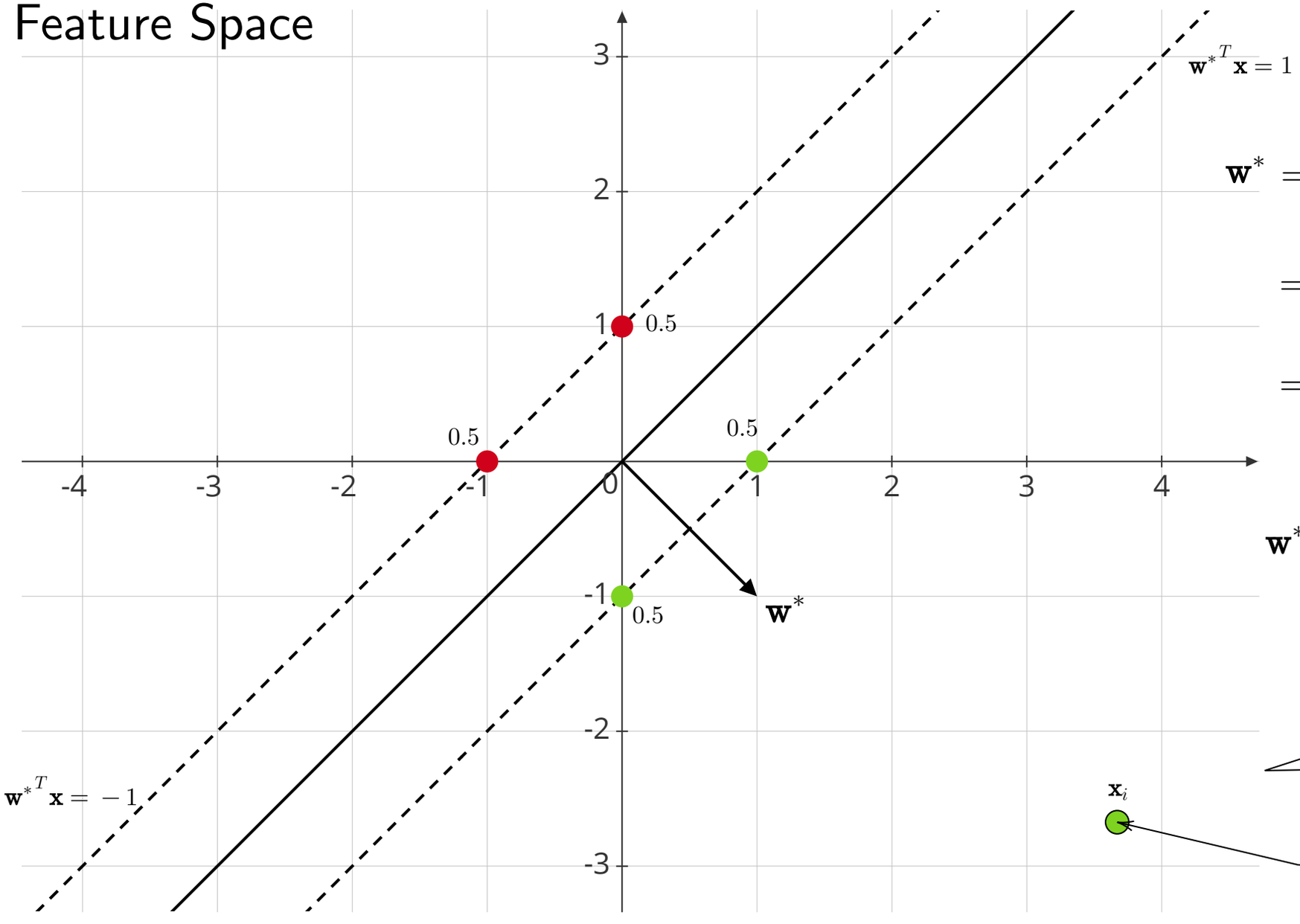
$$\mathbf{w}^{*T}\mathbf{x}_{\text{test}} = \sum_{i=1}^{n} \alpha_i \cdot \underbrace{\left[(y_i \mathbf{x}_i)^T \mathbf{x}_{\text{test}} \right]}_{\text{similarity}}$$





Feature Space

Example-1

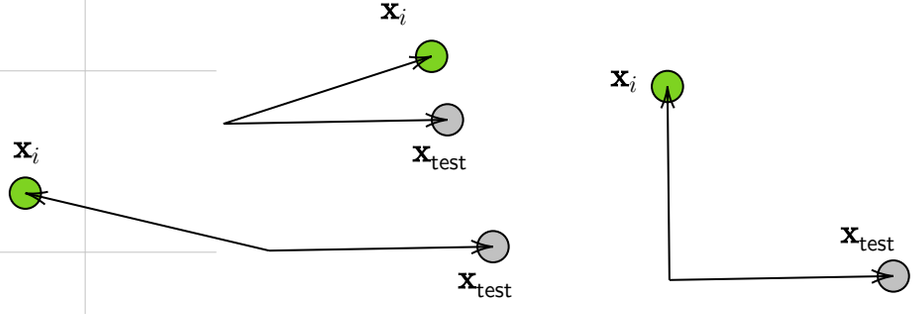


$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

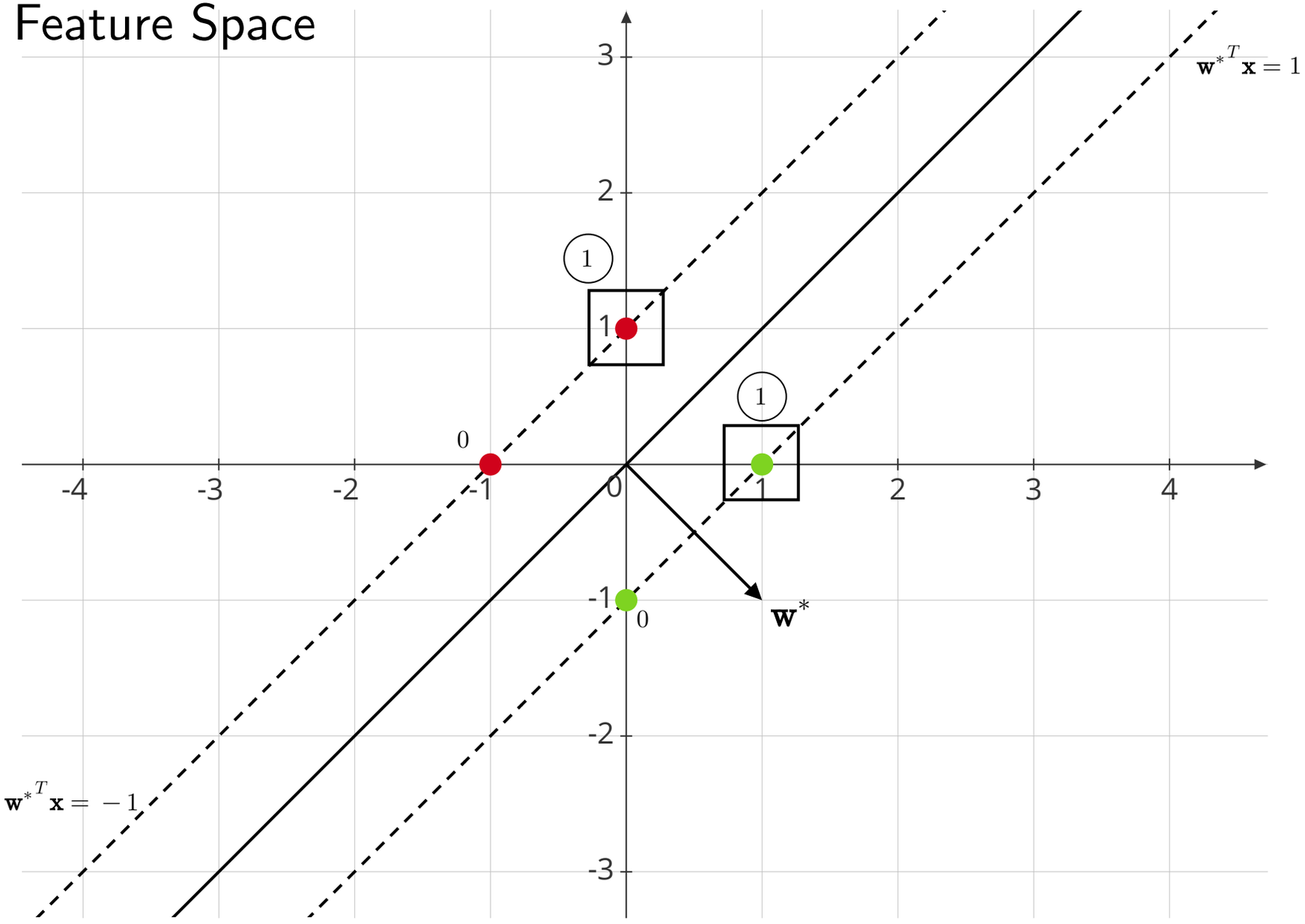
$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\mathbf{w}^{*T}\mathbf{x}_{\text{test}} = \sum_{i=1}^{n} \alpha_i \cdot \underbrace{\left[(y_i \mathbf{x}_i)^T \mathbf{x}_{\text{test}} \right]}_{\text{similarity}}$$



Feature Space



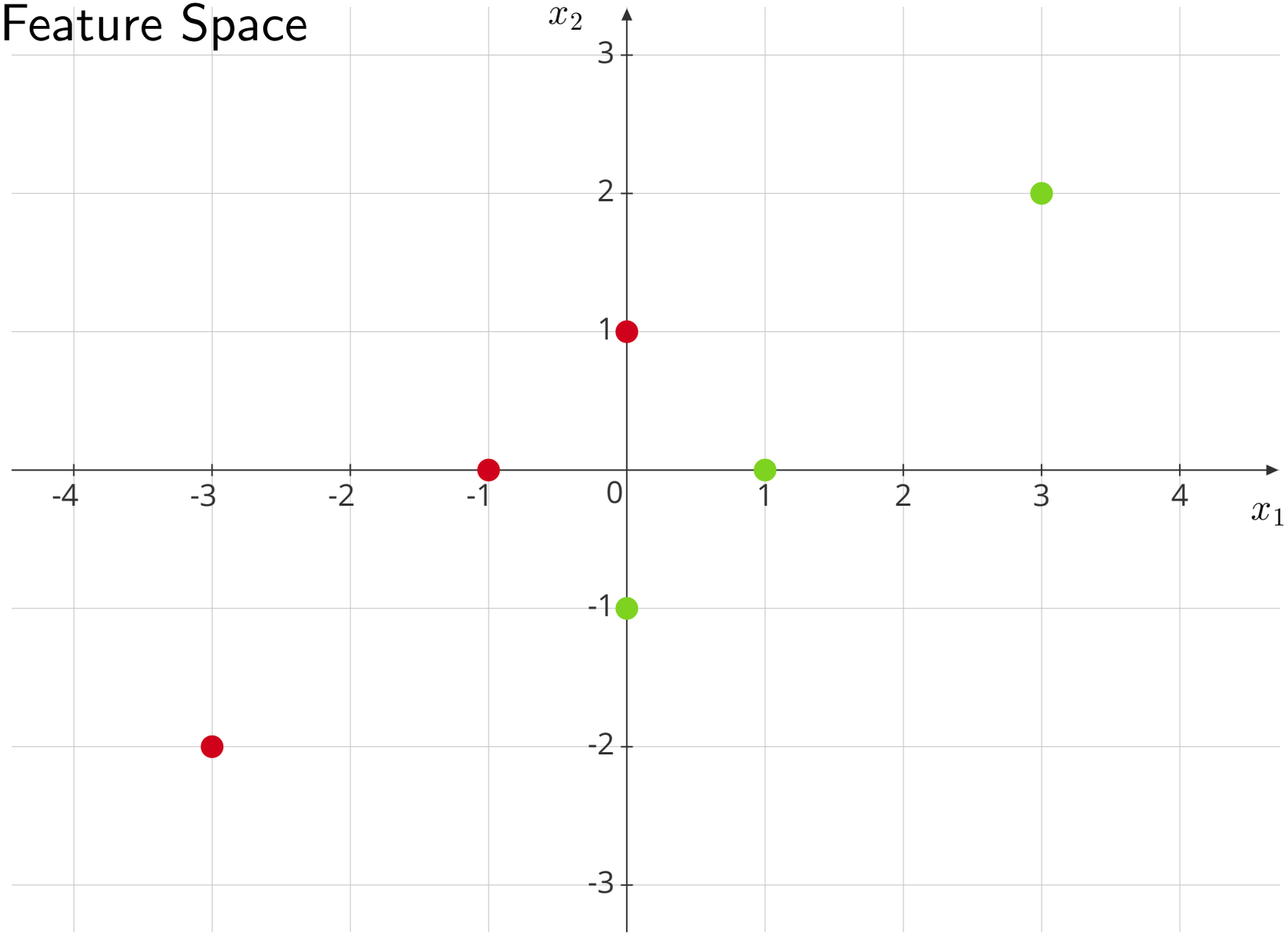
Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\boldsymbol{\alpha}^* = \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix}^T$$

$$\begin{aligned} \mathbf{w}^* &= \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i \\ &= 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ -1 \end{bmatrix} \end{aligned}$$

Feature Space

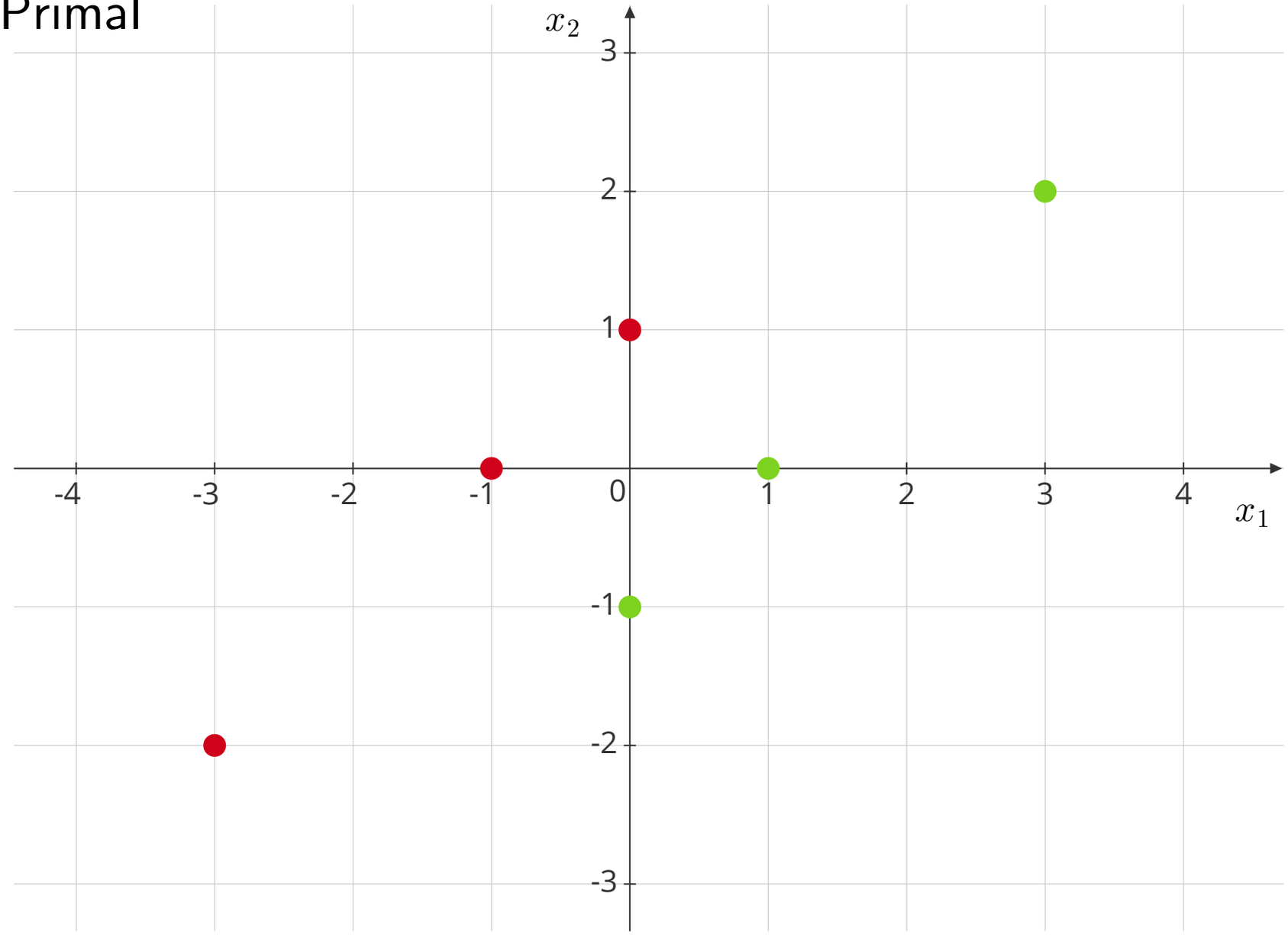


Example-2

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & -1 & 0 & -3 \\ 0 & -1 & 2 & 0 & 1 & -2 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \\ -1 \\ -1 \end{bmatrix}$$

Primal



Example-2

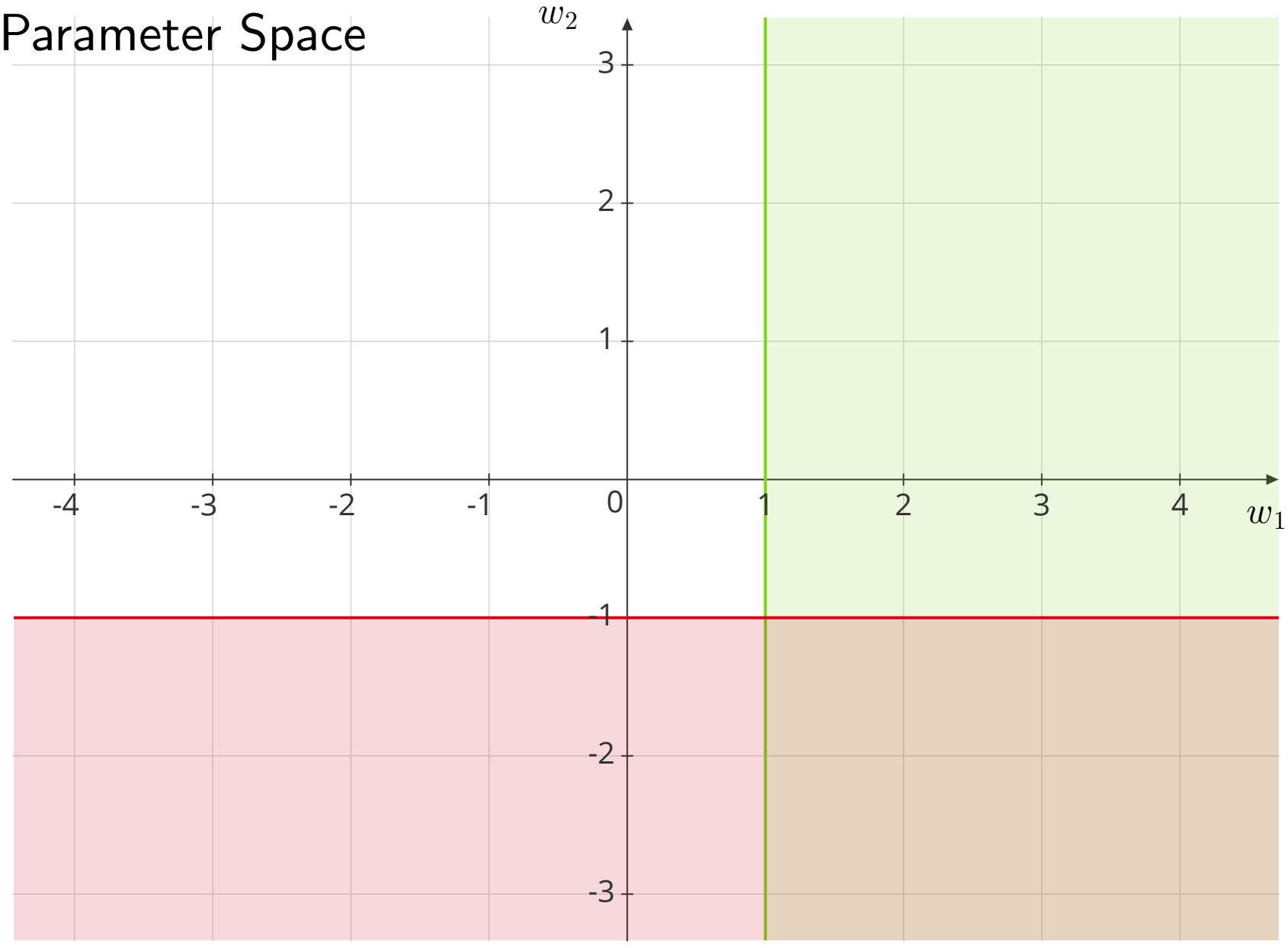
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & -1 & 0 & -3 \\ 0 & -1 & 2 & 0 & 1 & -2 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1 && (4) \\ w_2 &\leqslant -1 && (5) \\ 3w_1 + 2w_2 &\geqslant 1 && (6) \end{aligned}$$

Parameter Space



Example-2

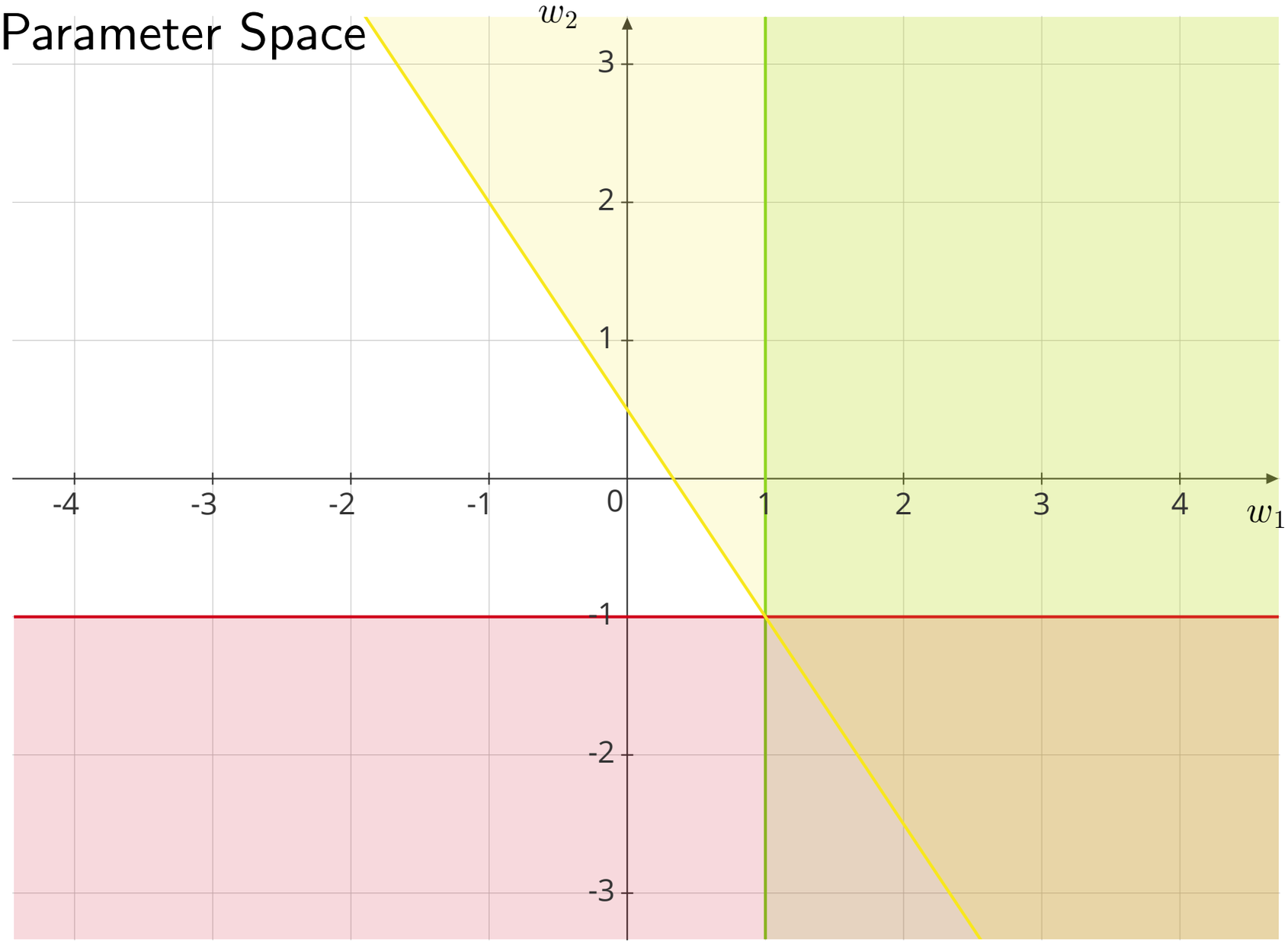
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & -1 & 0 & -3 \\ 0 & -1 & 2 & 0 & 1 & -2 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1 && (4) \\ w_2 &\leqslant -1 && (5) \\ 3w_1 + 2w_2 &\geqslant 1 && (6) \end{aligned}$$

Parameter Space



Example-2

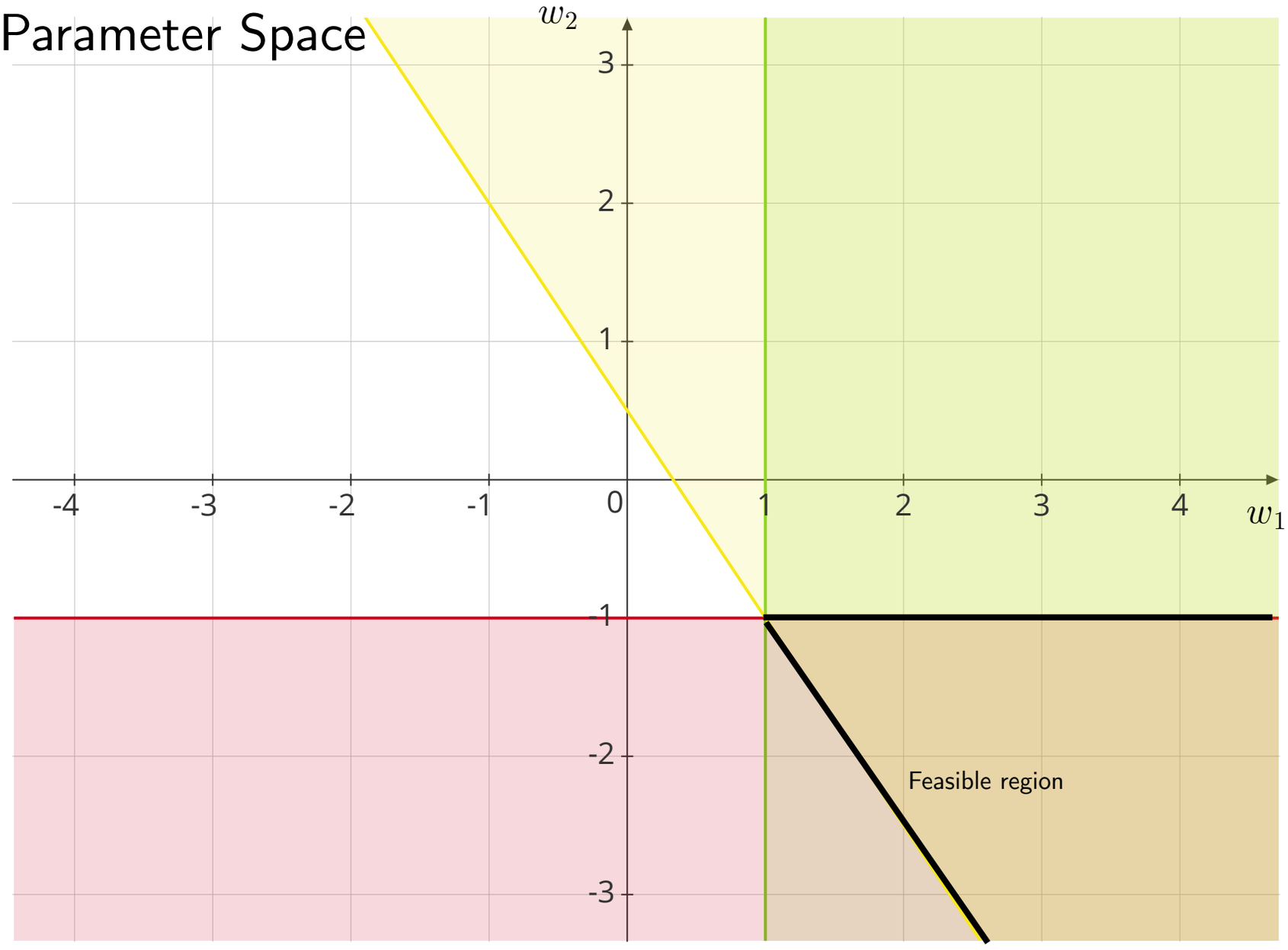
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & -1 & 0 & -3 \\ 0 & -1 & 2 & 0 & 1 & -2 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1 && (4) \\ w_2 &\leqslant -1 && (5) \\ 3w_1 + 2w_2 &\geqslant 1 && (6) \end{aligned}$$

Parameter Space



Example-2

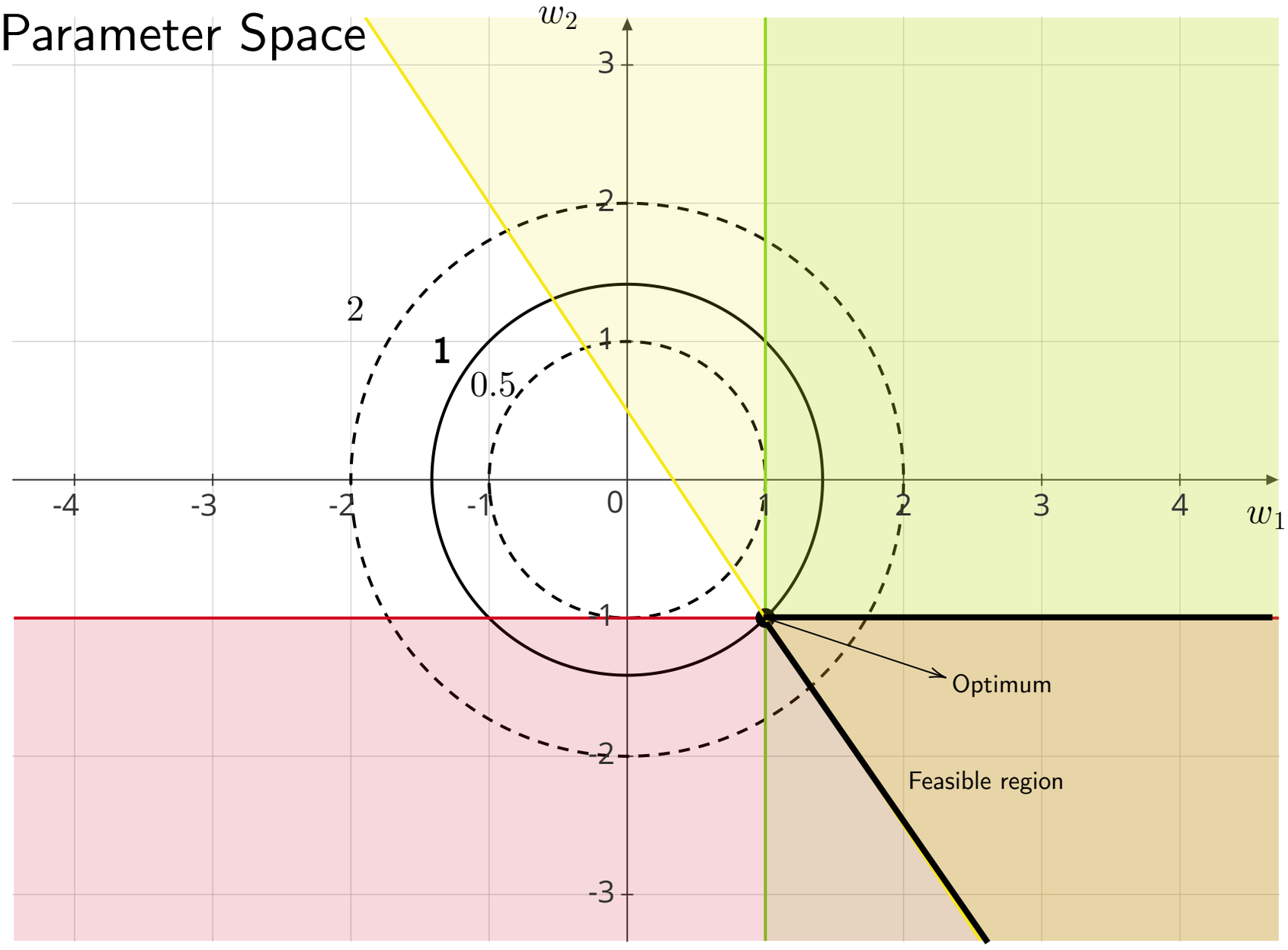
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & -1 & 0 & -3 \\ 0 & -1 & 2 & 0 & 1 & -2 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1 && (4) \\ w_2 &\leqslant -1 && (5) \\ 3w_1 + 2w_2 &\geqslant 1 && (6) \end{aligned}$$

Parameter Space



Example-2

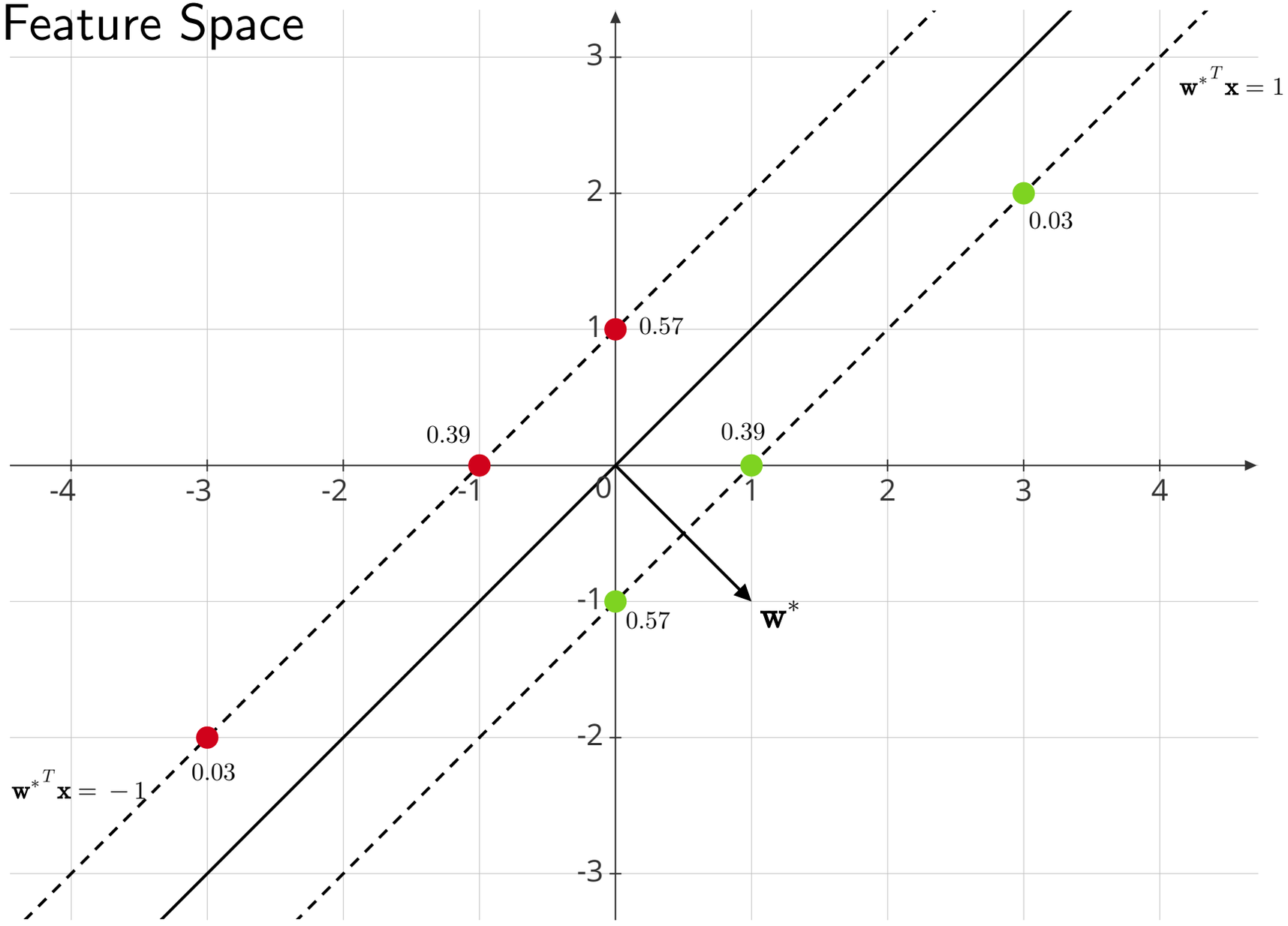
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & -1 & 0 & -3 \\ 0 & -1 & 2 & 0 & 1 & -2 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1 && (4) \\ w_2 &\leqslant -1 && (5) \\ 3w_1 + 2w_2 &\geqslant 1 && (6) \end{aligned}$$

Feature Space



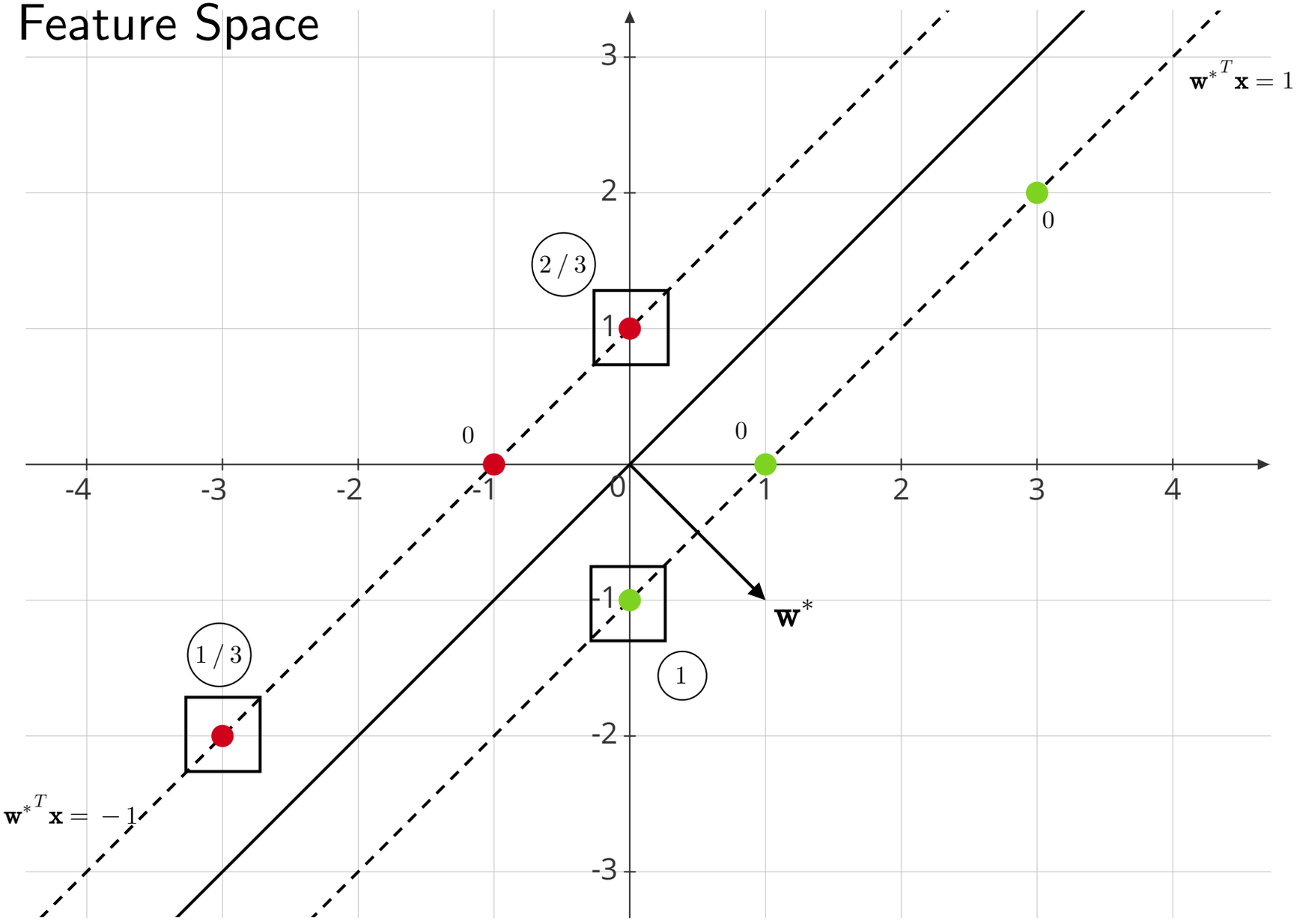
Example-2

$$\boldsymbol{\alpha}^* = \begin{bmatrix} 0.39 \\ 0.57 \\ 0.03 \\ 0.39 \\ 0.57 \\ 0.03 \end{bmatrix}$$

$$\mathbf{w}^* = \sum_{i=1}^{6} \alpha_i^* \mathbf{x}_i y_i$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space



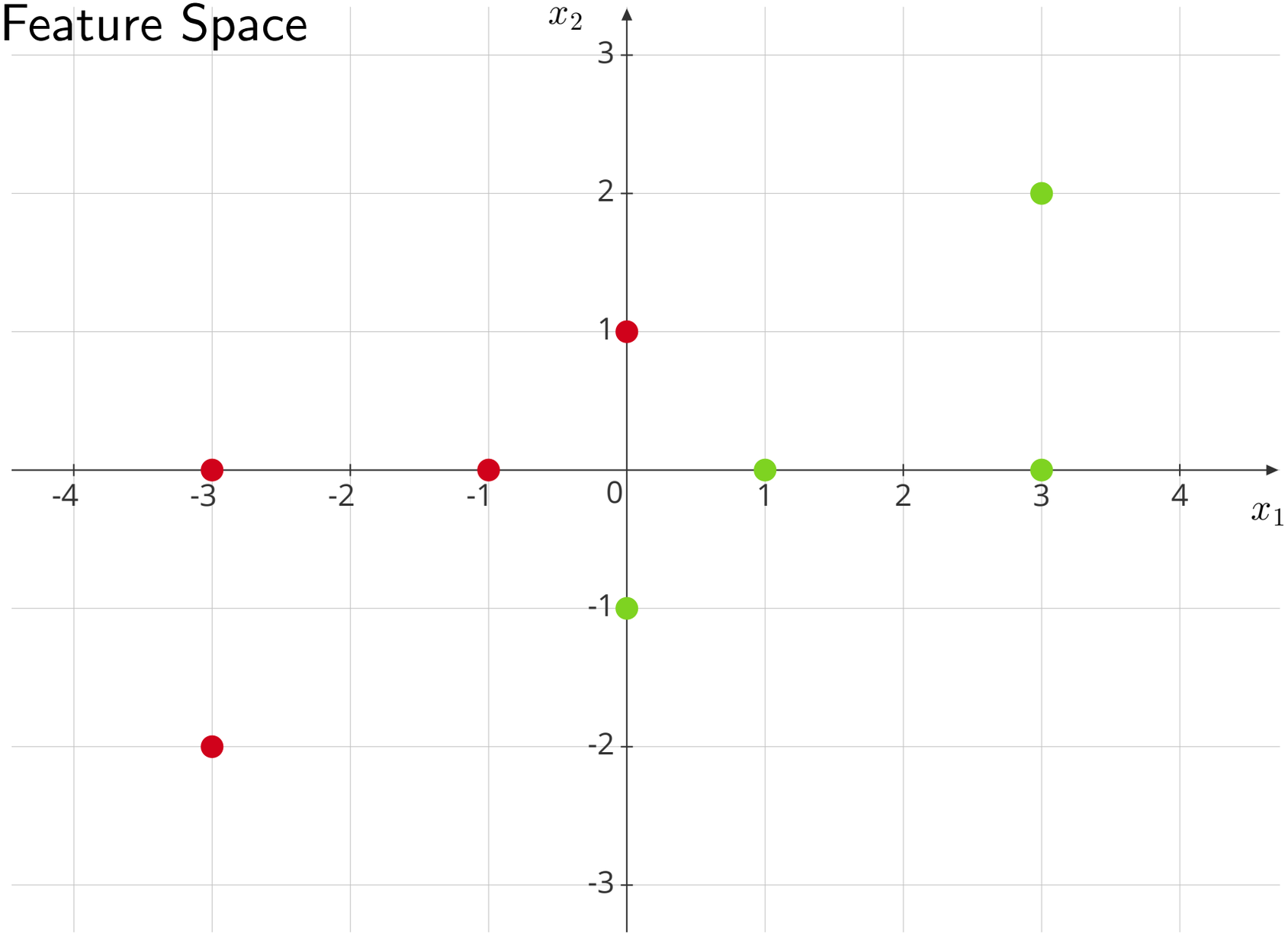
Example-2

$$\boldsymbol{\alpha}^* = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 2/3 \\ 1/3 \end{bmatrix}$$

$$\mathbf{w}^* = \sum_{i=1}^{6} \alpha_i^* \mathbf{x}_i y_i$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space

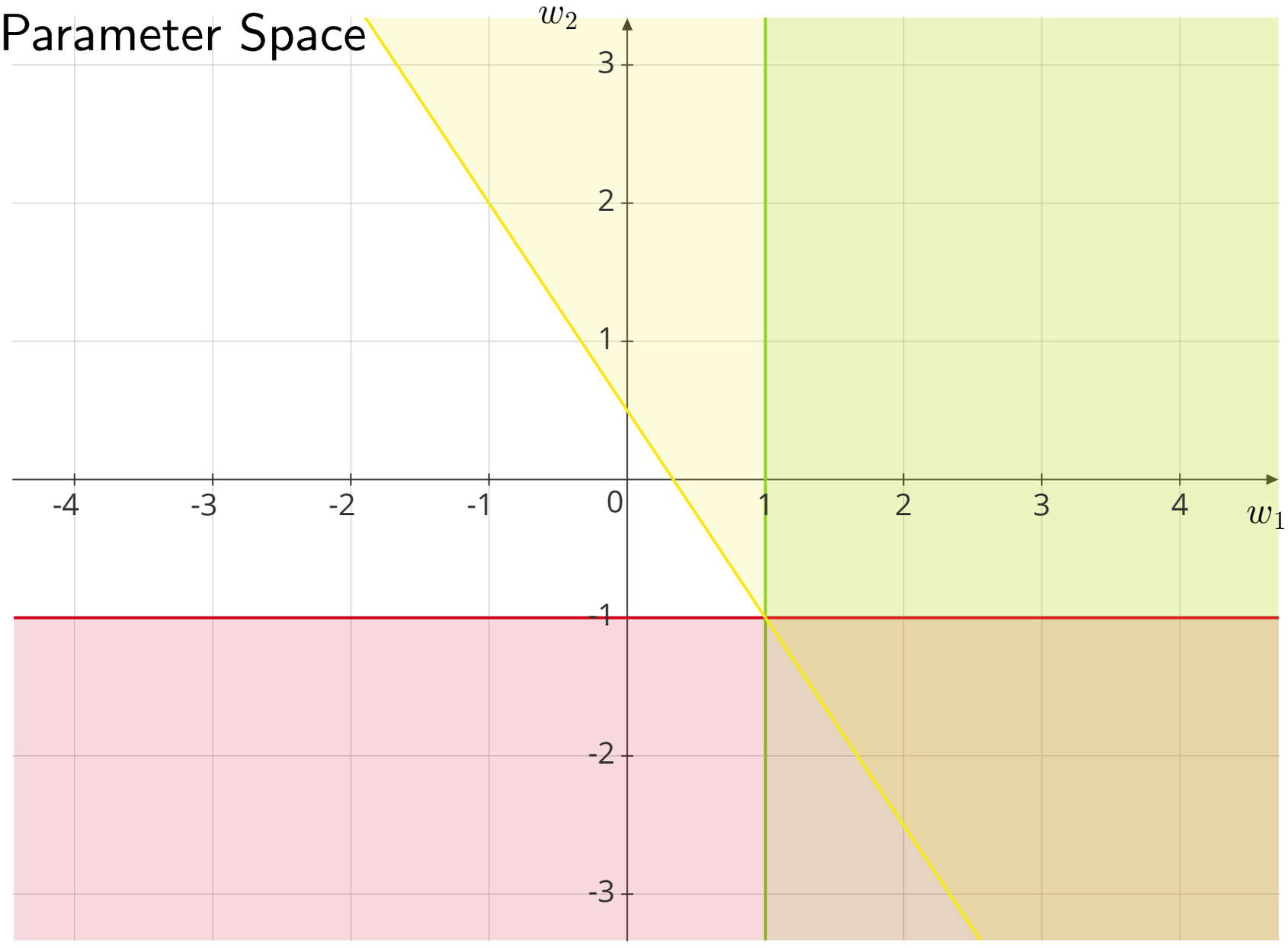


Example-3

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & 3 & -1 & 0 & -3 & -3 \\ 0 & -1 & 2 & 0 & 0 & 1 & -2 & 0 \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix}$$

Parameter Space



Example-3

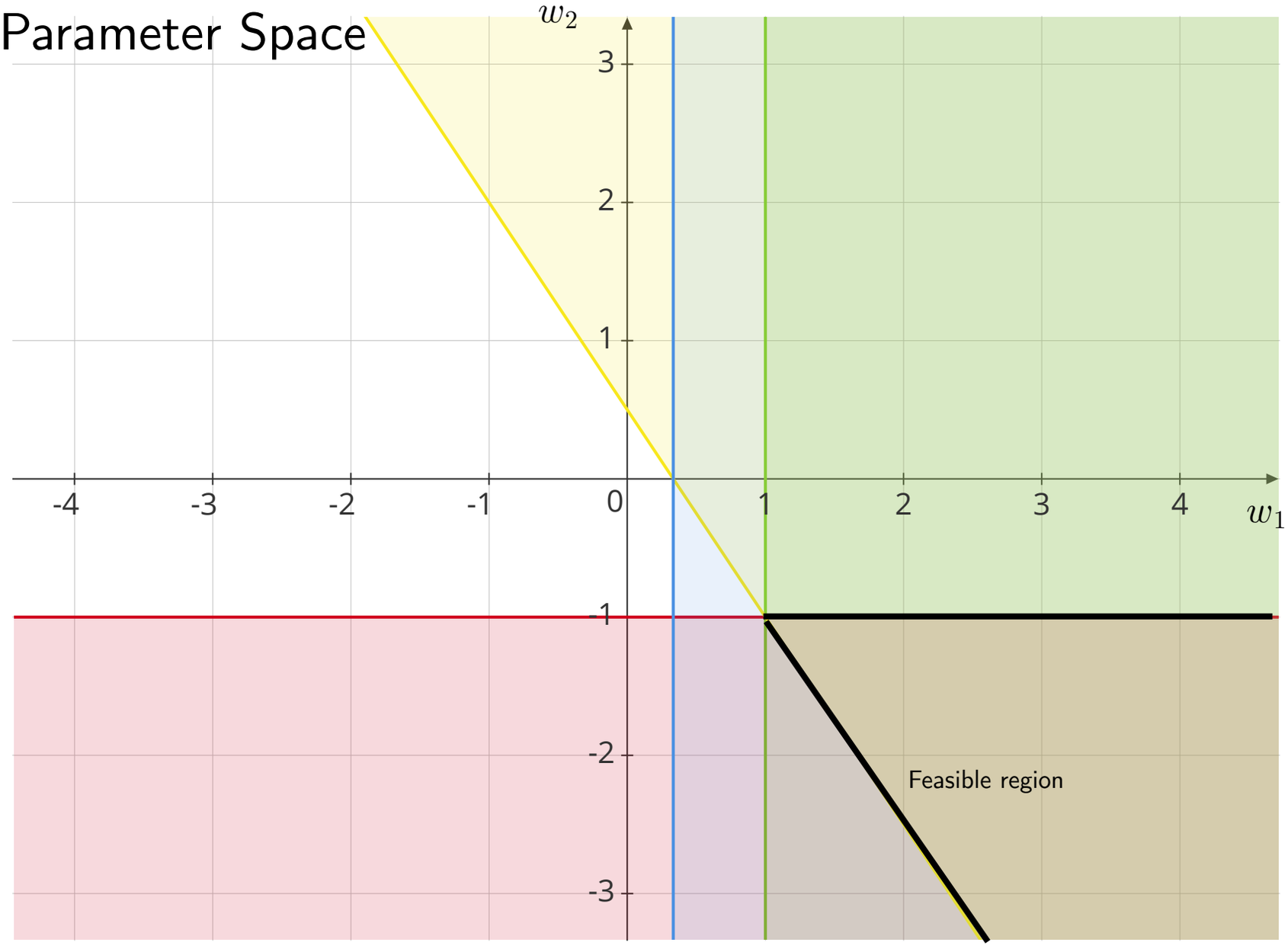
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & 3 & -1 & 0 & -3 & -3 \\ 0 & -1 & 2 & 0 & 0 & 1 & -2 & 0 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1/3 && (4) \\ w_1 &\geqslant 1 && (5) \\ w_2 &\leqslant -1 && (6) \\ 3w_1 + 2w_2 &\geqslant 1 && (7) \\ w_1 &\geqslant 1/3 && (8) \end{aligned}$$

Parameter Space



Example-3

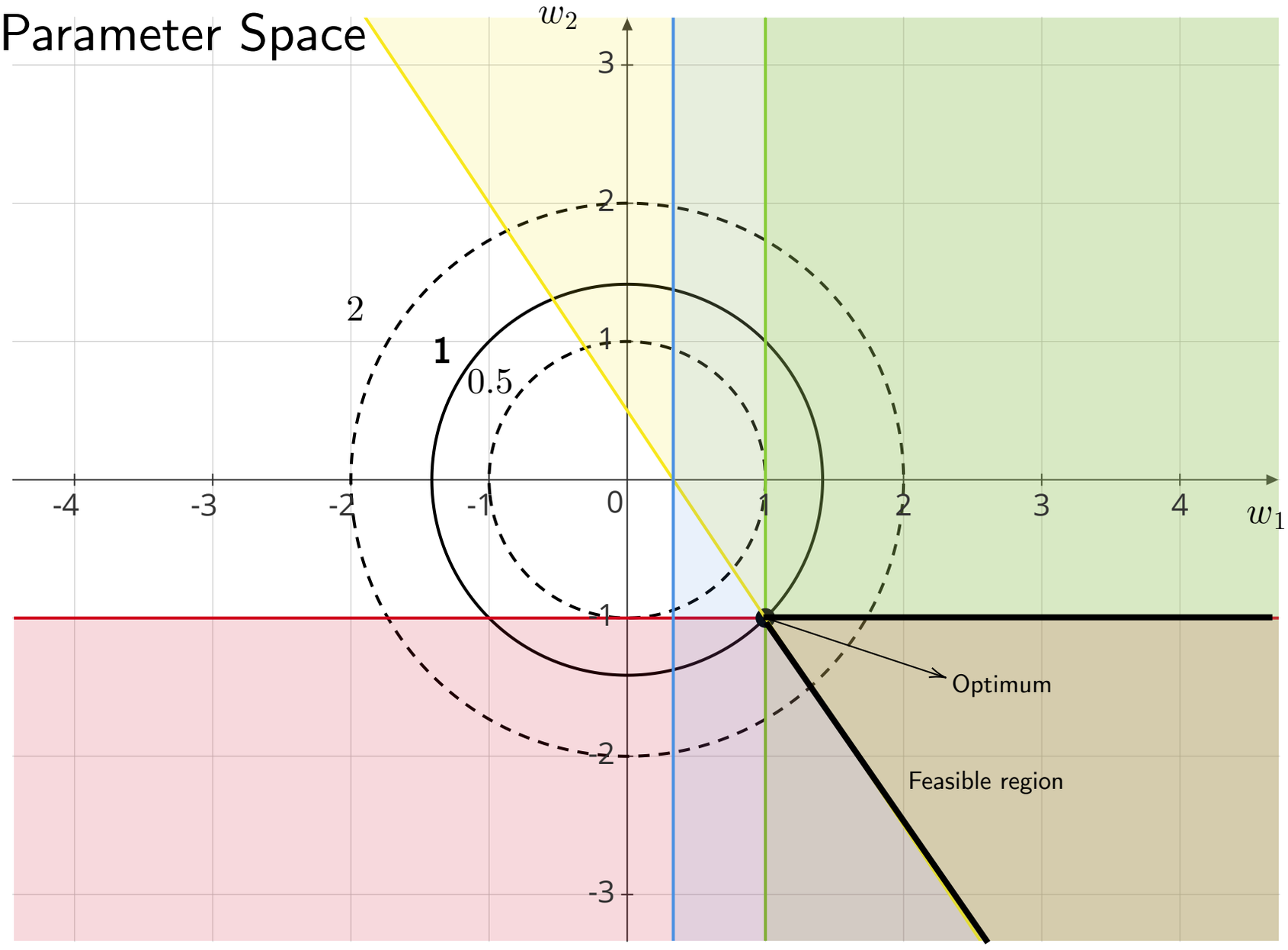
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & 3 & -1 & 0 & -3 & -3 \\ 0 & -1 & 2 & 0 & 0 & 1 & -2 & 0 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1/3 && (4) \\ w_1 &\geqslant 1 && (5) \\ w_2 &\leqslant -1 && (6) \\ 3w_1 + 2w_2 &\geqslant 1 && (7) \\ w_1 &\geqslant 1/3 && (8) \end{aligned}$$

Parameter Space



Example-3

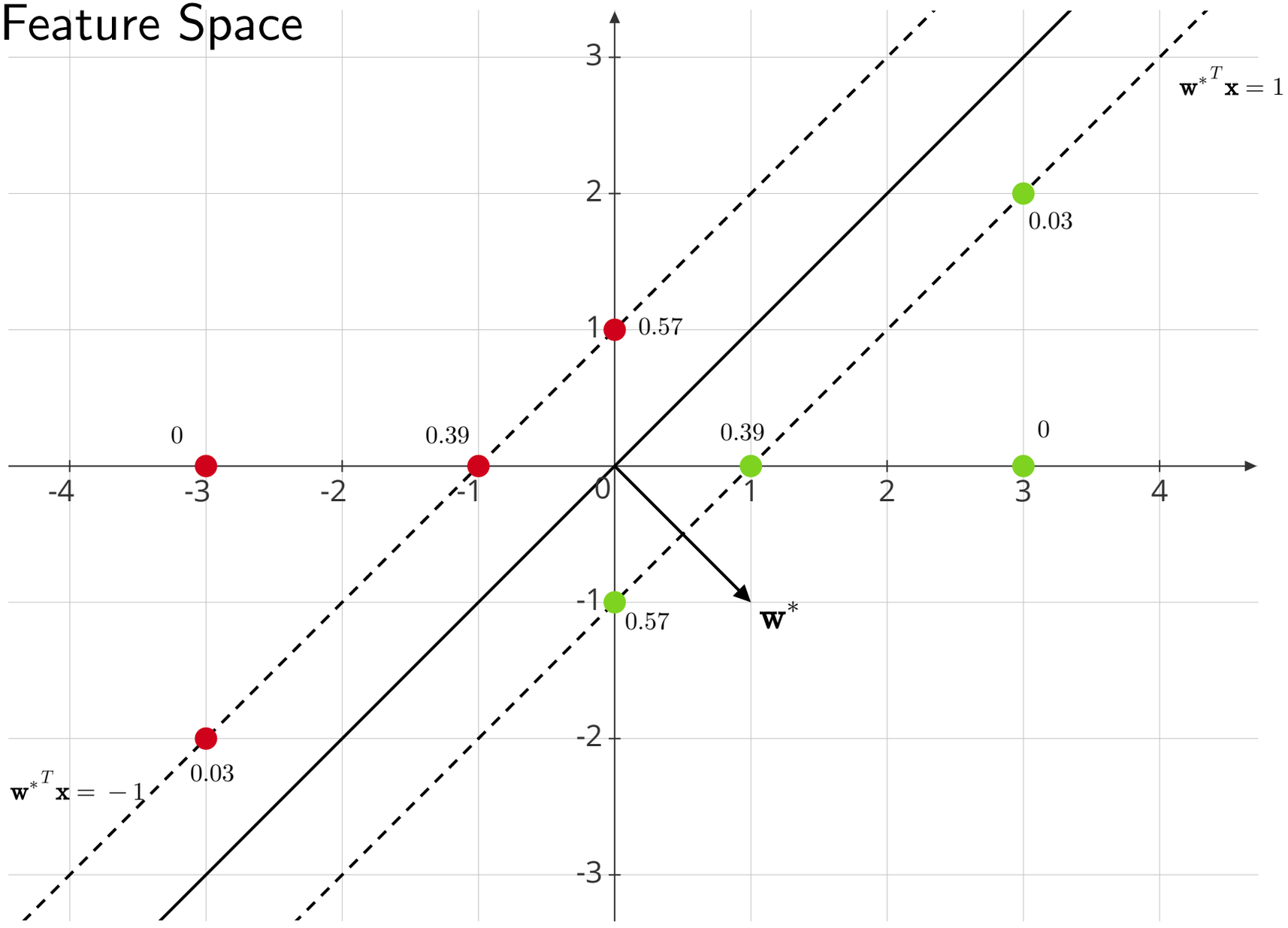
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & 3 & -1 & 0 & -3 & -3 \\ 0 & -1 & 2 & 0 & 0 & 1 & -2 & 0 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1/3 && (4) \\ w_1 &\geqslant 1 && (5) \\ w_2 &\leqslant -1 && (6) \\ 3w_1 + 2w_2 &\geqslant 1 && (7) \\ w_1 &\geqslant 1/3 && (8) \end{aligned}$$

Feature Space



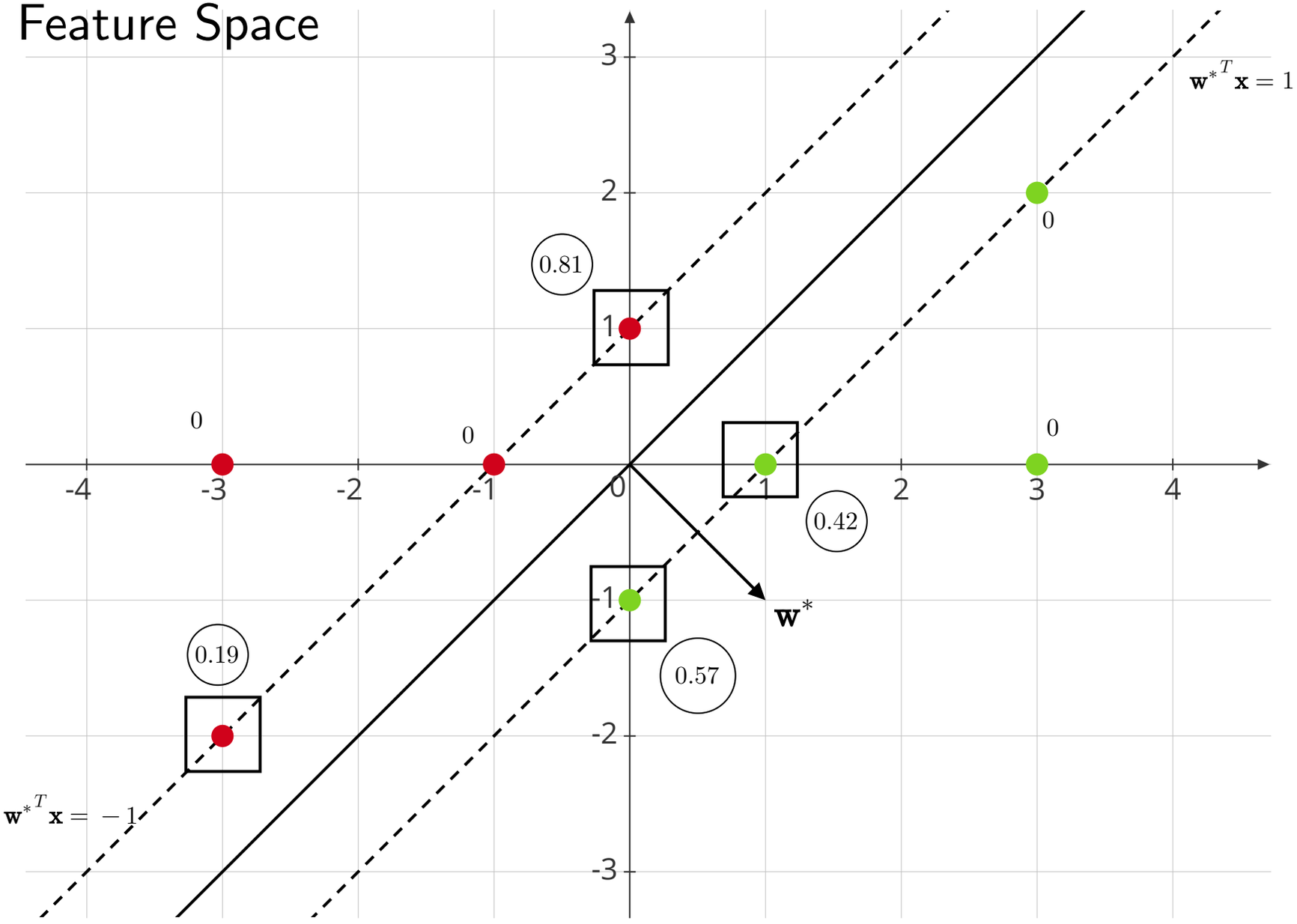
Example-2

$$\boldsymbol{\alpha}^* = \begin{bmatrix} 0.39 \\ 0.57 \\ 0.03 \\ 0 \\ 0.39 \\ 0.57 \\ 0.03 \\ 0 \end{bmatrix}$$

$$\mathbf{w}^* = \sum_{i=1}^{6} \alpha_i^* \mathbf{x}_i y_i$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space



Example-3

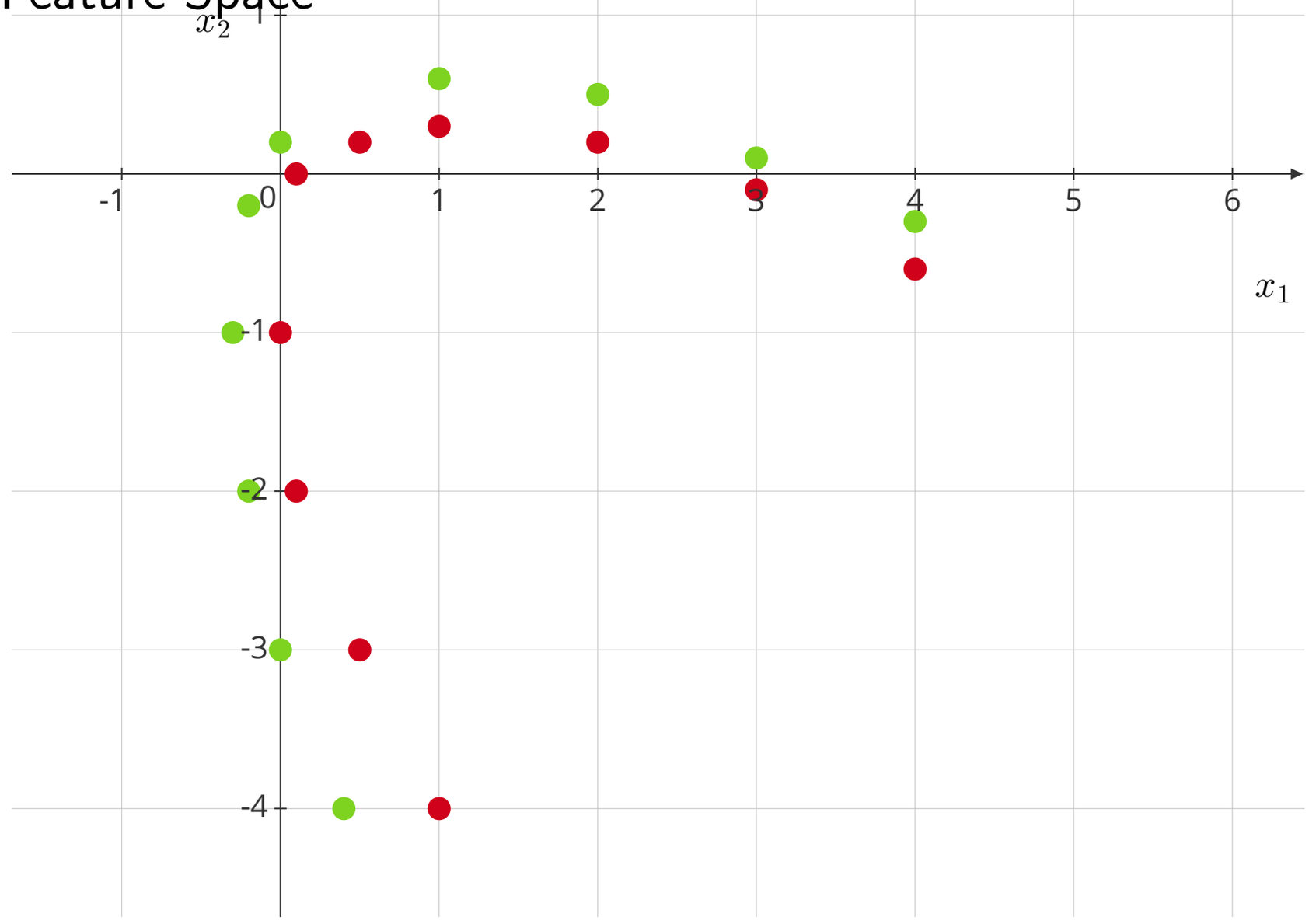
$$\boldsymbol{\alpha}^* = \begin{bmatrix} 0.42 \\ 0.57 \\ 0 \\ 0 \\ 0 \\ 0.81 \\ 0.19 \\ 0 \end{bmatrix}$$

$$\mathbf{w}^* = \sum_{i=1}^{6} \alpha_i^* \mathbf{x}_i y_i$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space

Example-4



Linear-SVM

$$\mathbf{x}_i$$

$$\mathbf{X}$$

$$\mathbf{X}^T\mathbf{X}$$

$$\max_{\boldsymbol{\alpha}\geqslant 0} \ \boldsymbol{\alpha}^T\mathbf{1} - \frac{1}{2}\cdot\boldsymbol{\alpha}^T\mathbf{Y}^T\mathbf{X}^T\mathbf{X}\mathbf{Y}\boldsymbol{\alpha}$$

$$\mathbf{w}^* = \sum_{i=1}^{n} \alpha_i^* \cdot \mathbf{x}_i \cdot y_i$$

$$\mathbf{w}^{*T}\mathbf{x}_{\text{test}} = \sum_{i=1}^{n} \alpha_i \cdot \underbrace{\left[(y_i\mathbf{x}_i)^T\mathbf{x}_{\text{test}}\right]}_{\text{similarity}}$$

Kernel-SVM

Linear-SVM

$$\mathbf{x}_i$$

$$\mathbf{X}$$

$$\mathbf{X}^T\mathbf{X}$$

$$\max_{\boldsymbol{\alpha}\geqslant 0} \ \boldsymbol{\alpha}^T\mathbf{1} - \frac{1}{2}\cdot\boldsymbol{\alpha}^T\mathbf{Y}^T\mathbf{X}^T\mathbf{X}\mathbf{Y}\boldsymbol{\alpha}$$

$$\mathbf{w}^* = \sum_{i=1}^{n} \alpha_i^* \cdot \mathbf{x}_i \cdot y_i$$

$$\mathbf{w}^{*T}\mathbf{x}_{\text{test}} = \sum_{i=1}^{n} \alpha_i \cdot \underbrace{\left[(y_i\mathbf{x}_i)^T\mathbf{x}_{\text{test}}\right]}_{\text{similarity}}$$

Kernel-SVM

$$\phi(\mathbf{x}_i)$$

$$\phi(\mathbf{X})$$

$$\mathbf{K}$$

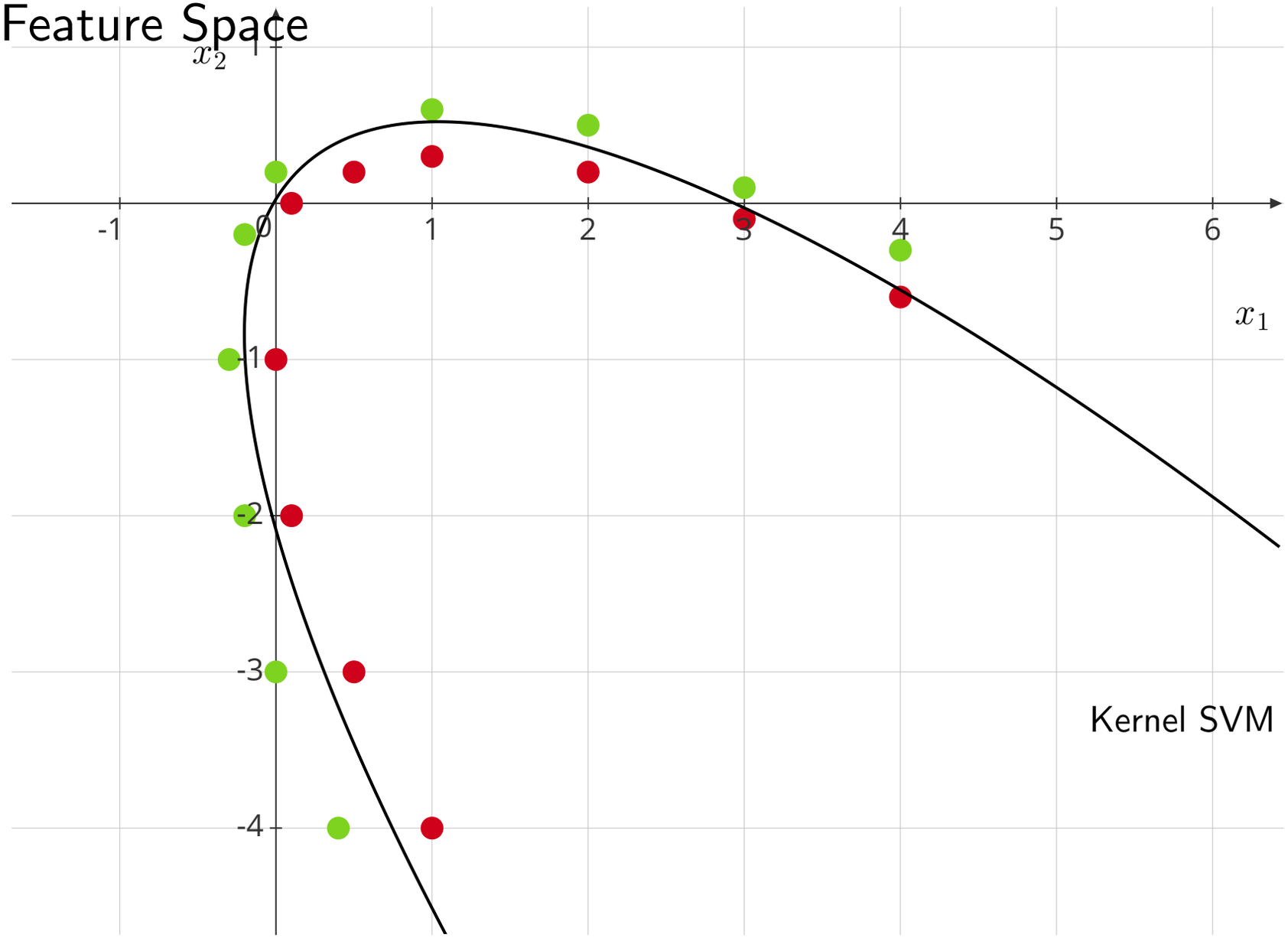
$$\max_{\boldsymbol{\alpha}\geqslant 0} \ \boldsymbol{\alpha}^T\mathbf{1} - \frac{1}{2}\cdot\boldsymbol{\alpha}^T\mathbf{Y}^T\mathbf{K}\mathbf{Y}\boldsymbol{\alpha}$$

$$\mathbf{w}^* = \sum_{i=1}^{n} \alpha_i^* \cdot \phi(\mathbf{x}_i) \cdot y_i$$

$$\mathbf{w}^{*T}\mathbf{x}_{\text{test}} = \sum_{i=1}^{n} \alpha_i \cdot \underbrace{y_i \cdot k(\mathbf{x}_i, \mathbf{x}_{\text{test}})}_{\text{similarity}}$$

Feature Space

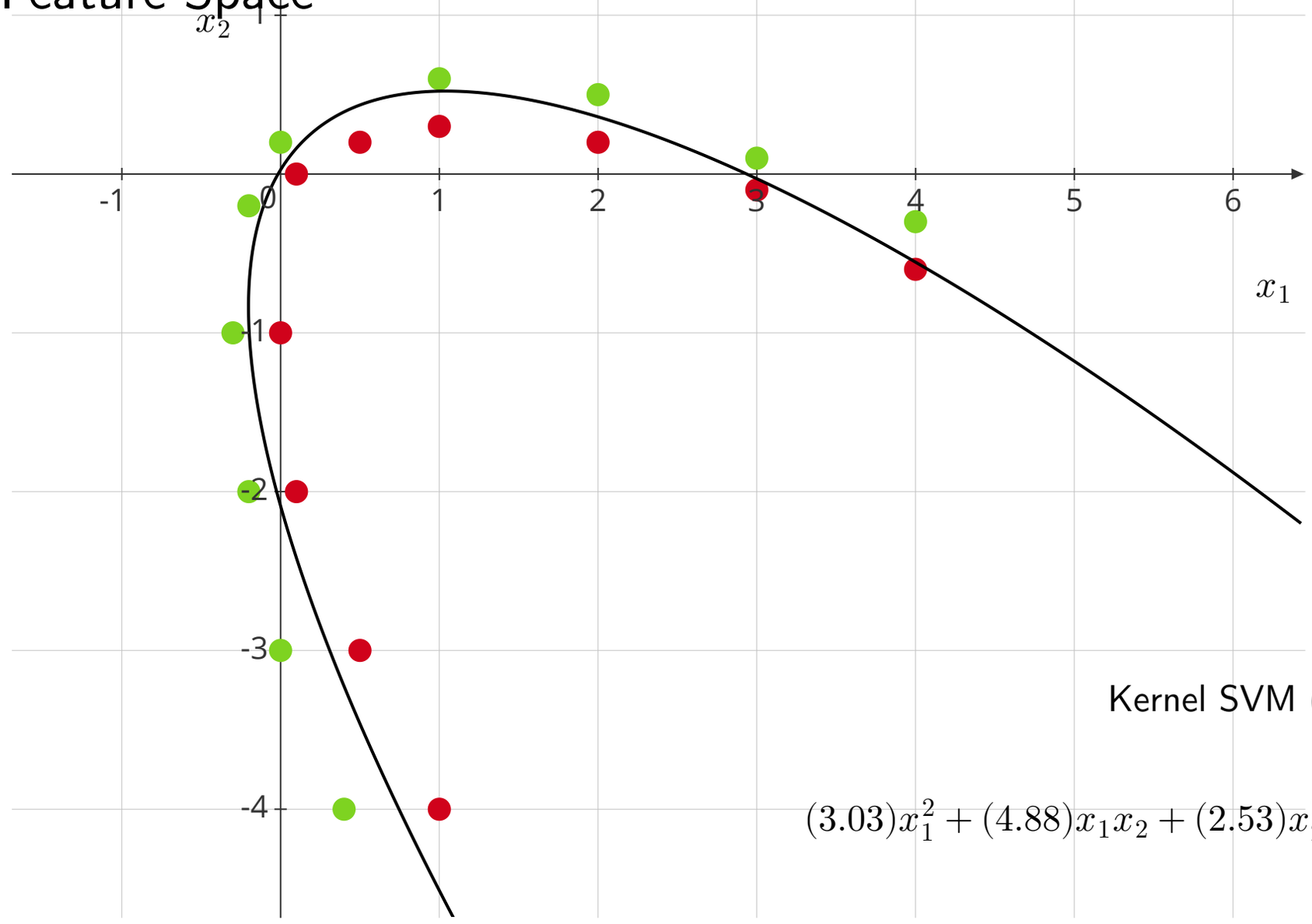
Example-4



Kernel SVM (quadratic)

Feature Space

Example-4

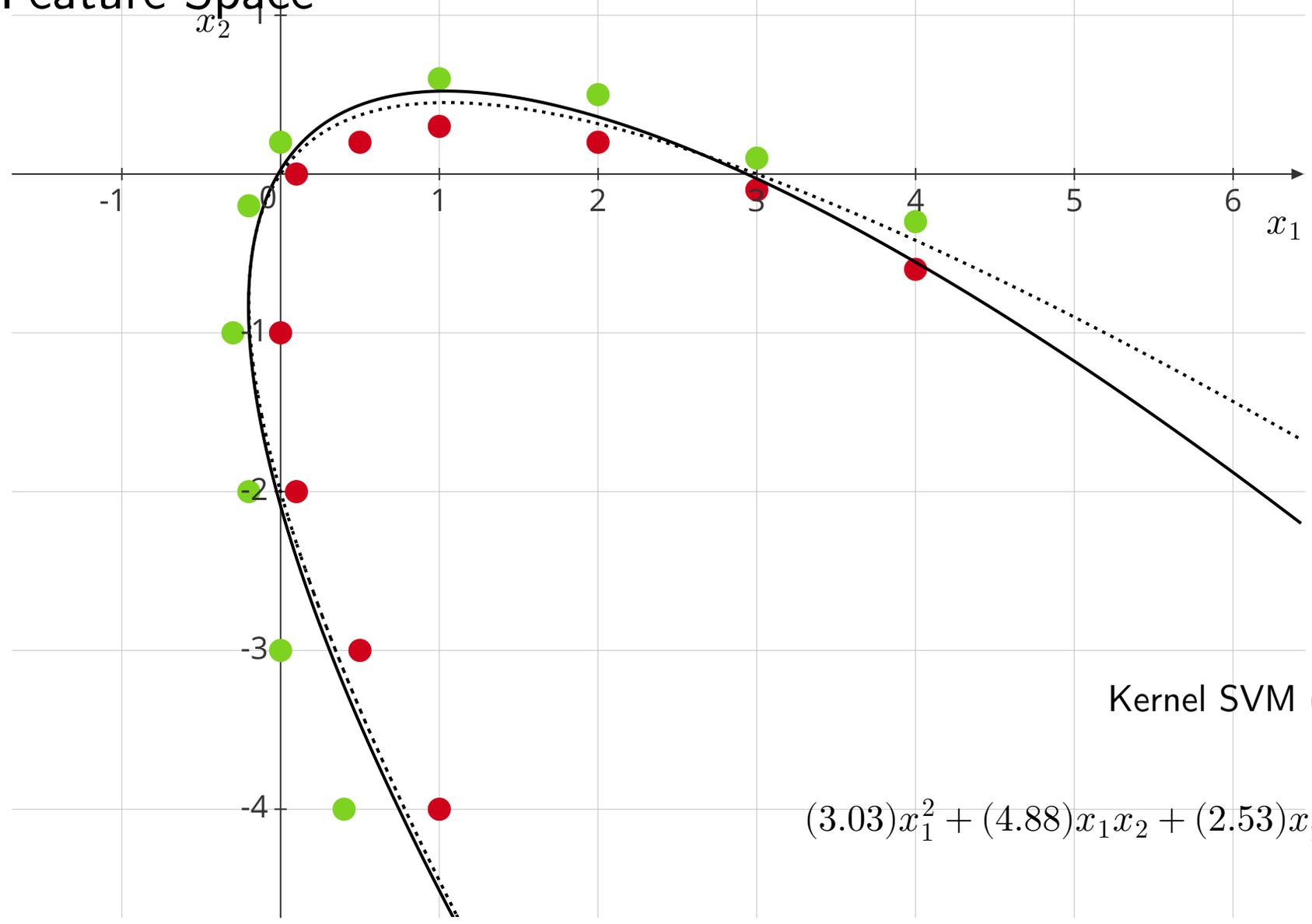


Kernel SVM (quadratic)

$$(3.03)x_1^2 + (4.88)x_1x_2 + (2.53)x_2^2 - (8.85)x_1 + (5.22)x_2 - (0.14) = 0$$

Feature Space

Example-4



Source

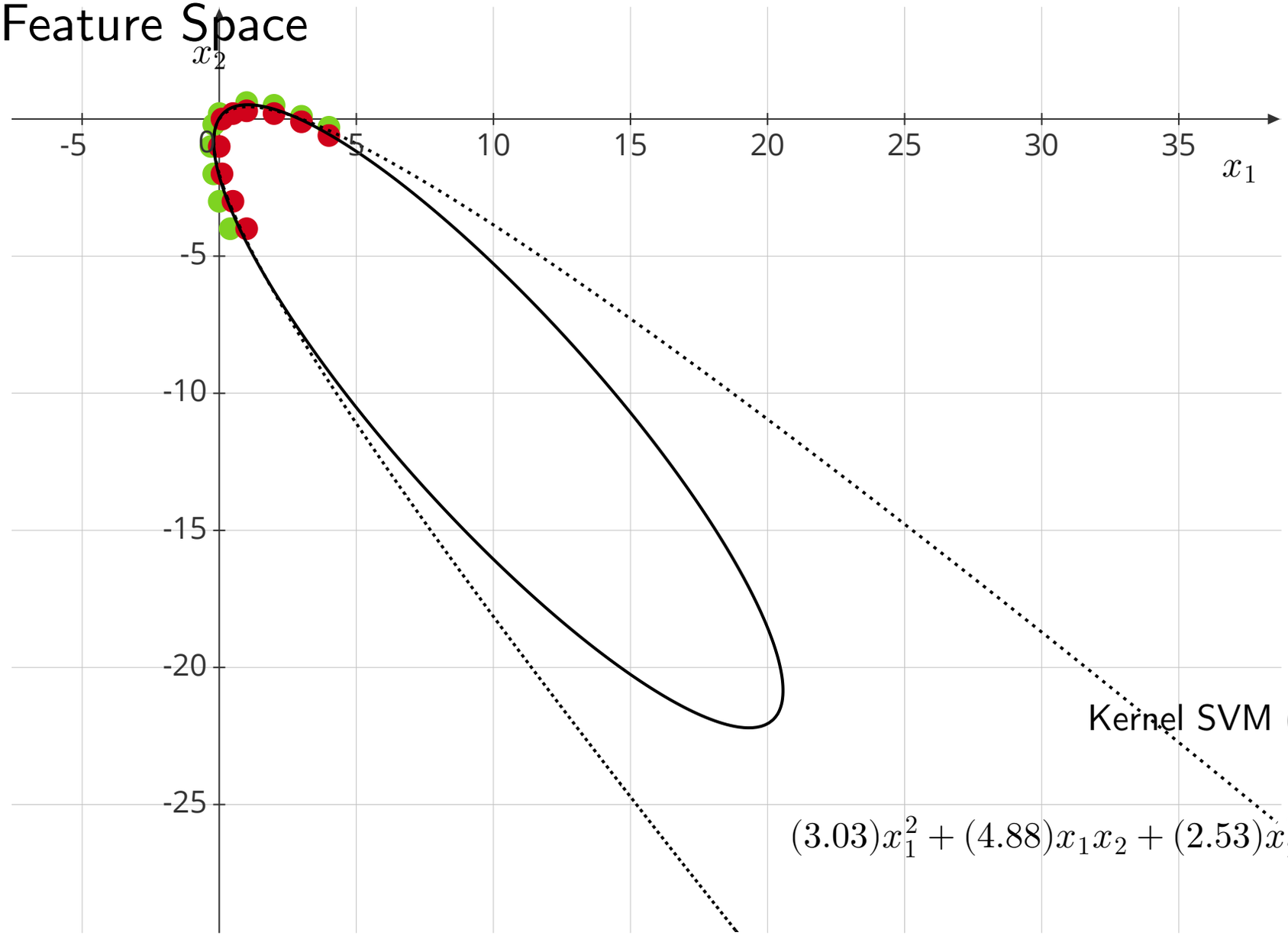
$$x_1^2 + 2x_1x_2 + x_2^2 - 3x_1 + 2x_2 = 0$$

Kernel SVM (quadratic)

$$(3.03)x_1^2 + (4.88)x_1x_2 + (2.53)x_2^2 - (8.85)x_1 + (5.22)x_2 - (0.14) = 0$$

Feature Space

Example-4



Source

$$x_1^2 + 2x_1x_2 + x_2^2 - 3x_1 + 2x_2 = 0$$

Kernel SVM (quadratic)

$$(3.03)x_1^2 + (4.88)x_1x_2 + (2.53)x_2^2 - (8.85)x_1 + (5.22)x_2 - (0.14) = 0$$